

Next-Generation Quantum Security: The Impact of Quantum Computing on Cybersecurity—Threats, Mitigations, and Solutions

Sijjad Ali ^{a,*,}, Shuaib Ahmed Wadho ^b, Kazim Raza Talpur ^c, Bandeh Ali Talpur ^d,
Khulud Salem Alshudukhi ^e, Mamoona Humayun ^f, Samar Raza Talpur ^g,
Mohd Abdullah Al Mamun ^h, Muhammad Naseem ⁱ, Adeel Abro ^j, Dhani Bux Talpur ^k,
Asadullah Shah ^k

^a College of Computer Science and Software Engineering, Shenzhen University, Guangdong, China

^b Faculty of Information and Communication Technology, Universiti Tunku Abdul Rahman, Kampar, Malaysia

^c Department of Intelligence Informatics, Faculty of Artificial Intelligence, Universiti Teknologi Malaysia, Kuala Lumpur, Malaysia

^d School of Computer Science and Statistics, Trinity College Dublin, Dublin, Ireland

^e Department of Computer Science, College of Computer and Information Sciences, Jouf University, Sakaka, Saudi Arabia

^f Department of Computing, Engineering and the Built Environment, University of Roehampton, London, United Kingdom

^g Sukkur IBA University, Sindh, Pakistan

^h MBA in Information Technology Management, Westcliff University, USA

ⁱ College of Media and Communication, Shenzhen University, Guangdong, China

^j Shaheed Allah Buksh Soomro University of Art, Design, and Heritages, SABS University, Jamshoro, Pakistan

^k Kulliyah of Information and Communication Technology, International Islamic University Malaysia, Malaysia

ARTICLE INFO

Keywords:

Quantum computing
Cybersecurity
Post quantum cryptography (PQC)
Quantum key distribution (QKD)
Quantum-safe cryptography
Quantum threats

ABSTRACT

The rapid advancement of quantum computing poses unprecedented threats to classical cryptographic systems, jeopardizing the security of digital infrastructures worldwide. This review paper comprehensively examines the transformative impact of quantum computing on cybersecurity, highlighting vulnerabilities introduced by algorithms like Shor's and Grover's, which can break widely used encryption methods such as RSA, ECC, and AES. It underscores the urgency of adopting quantum-resistant solutions, including post-quantum cryptography (PQC) and quantum key distribution (QKD), to mitigate these risks. The paper also explores emerging hybrid systems, quantum-safe blockchain, and IoT security, offering actionable insights for transitioning to quantum-resilient frameworks. By synthesizing interdisciplinary research, this work emphasizes the need for global collaboration, standardization, and innovation to safeguard data integrity and privacy in the impending quantum era, making it a pivotal resource for researchers, practitioners, and policymakers.

1. Introduction

Quantum computing represents a paradigm shift in computational science, leveraging the principles of quantum mechanics—such as superposition and entanglement to solve problems that are computationally infeasible for classical systems [1–4]. This

* Corresponding author.

E-mail addresses: sijjadali2021@email.szu.edu.cn (S. Ali), shuaib@lutar.my (S.A. Wadho), talpurkazim@utm.my (K.R. Talpur), talpurba@tcd.ie (B.A. Talpur), ksalshudukhi@ju.edu.sa (K.S. Alshudukhi), mamoona.humayun@roehampton.ac.uk (M. Humayun), samartalpur@gmail.com (S.R. Talpur), mamun.westcliffuniversity.usa@gmail.com (M.A.A. Mamun), Nk.achakzai@gmail.com (M. Naseem), dr.adeelabro@sabsu.edu.pk (A. Abro), ghanibux@usms.edu.pk (D.B. Talpur), asadullah@iium.edu.my (A. Shah).

<https://doi.org/10.1016/j.compeleceng.2025.110649>

Received 9 March 2025; Received in revised form 26 July 2025; Accepted 24 August 2025

Available online 31 August 2025

0045-7906/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

emerging technology introduces fundamentally new ways of processing information, enabling certain algorithms to outperform their classical counterparts by orders of magnitude. Among the most prominent examples are Shor's algorithm for factoring large integers and computing discrete logarithms [5], and Grover's algorithm for unstructured search [6], both of which have direct implications for the foundational structures of modern cryptography. The exponential speedup offered by Shor's algorithm threatens to render widely used public-key encryption schemes—such as RSA and elliptic curve cryptography (ECC) obsolete [7], while Grover's quadratic speedup reduces the effective security of symmetric-key algorithms, including AES [8,9]. These vulnerabilities could compromise the confidentiality, integrity, and authenticity of sensitive data across financial, governmental, military, and civilian digital infrastructures.

While the practical realization of large-scale, fault-tolerant quantum computers remains a significant technical challenge [10,11], progress in quantum hardware development suggests that such capabilities may be attainable within a few decades—or even sooner, given recent advances by leading technology firms and research institutions. This uncertain yet plausible timeline creates a strategic dilemma: cryptographic systems must be secured well in advance of quantum adversaries to prevent future decryption of intercepted data [12,13]. This concern is embodied in the harvest now, decrypt later paradigm, in which encrypted communications captured today could be retrospectively decrypted once quantum capabilities mature. This scenario is especially troubling for data with long-term confidentiality requirements, such as classified government documents, healthcare records, or corporate trade secrets.

In response to these emerging threats, two principal countermeasures have been proposed: post-quantum cryptography (PQC) and quantum key distribution (QKD). PQC involves the development of cryptographic algorithms based on mathematical problems that are believed to be resistant to both classical and quantum attacks [14], such as lattice-based, code-based, multivariate polynomial, and hash-based schemes. These algorithms are intended to serve as drop-in replacements for current cryptosystems and are being actively evaluated by standardization bodies such as the National Institute of Standards and Technology (NIST). QKD, in contrast, uses the principles of quantum mechanics to establish cryptographic keys with information-theoretic security, offering a fundamentally different approach to secure communication [15]. However, both PQC and QKD face critical implementation challenges. PQC must balance security with performance and compatibility with existing protocols, while QKD is constrained by practical limitations in transmission distance, hardware complexity, and integration into conventional networks.

Despite growing interest and funding in quantum-resilient technologies, the broader cybersecurity community remains fragmented in its response [16,17]. Industry adoption is slow, often hindered by uncertainty about the maturity and viability of proposed solutions, regulatory ambiguity, and the sheer complexity of retrofitting legacy systems. Furthermore, the absence of unified frameworks that bridge theoretical cryptographic advances with engineering feasibility and deployment readiness continues to impede meaningful progress [18,19]. While several academic and governmental initiatives have begun to address these concerns, comprehensive studies that holistically assess the landscape—incorporating cryptographic theory, hardware infrastructure, economic impact, regulatory considerations, and operational integration—are still lacking.

The motivation for our review is to address these multidimensional challenges by providing an in-depth, interdisciplinary examination of how quantum computing threatens existing cybersecurity paradigms and how emerging solutions can be effectively developed, standardized, and deployed. It critically evaluates the current state of PQC and QKD, explores hybrid models that combine classical and quantum techniques, and investigates the infrastructural, organizational, and geopolitical factors shaping the transition to quantum-safe systems. Through this synthesis, the review highlights the urgency of proactive and coordinated action, advocating for a global, cross-sectoral approach that fosters innovation, supports standardization efforts, and ensures a smooth and secure evolution of digital ecosystems. As the quantum era approaches, the stakes are too high for complacency; safeguarding information in a post-quantum world requires strategic foresight, sustained investment, and a collaborative spirit that transcends disciplinary and national boundaries.

The Fig. 1, outlines key threats posed by quantum computing to modern cryptography, including the potential break of RSA and ECC (elliptic curve cryptography) through Shor's algorithm, the weakening of hash functions and symmetric encryption via Grover's algorithm, and vulnerabilities in post-quantum cryptography, hash-based signatures, and multivariate polynomial systems. It also highlights risks in quantum key distribution (QKD), cloud security, and random number generation. To counter these threats, the figure points to “Emerging Solutions”, likely referring to ongoing advancements in post-quantum cryptographic algorithms (such as lattice-based or code-based cryptography), hybrid systems, and quantum-safe protocols designed to withstand quantum attacks. The structured layout effectively contrasts critical vulnerabilities with proactive mitigation strategies, emphasizing the need for a transition to quantum-resistant technologies.

In Table 1, we present a comprehensive overview of the impact of quantum computing on various aspects of cybersecurity, highlighting unique risks and opportunities, proposed mitigations, and open research questions. Each row addresses a specific cybersecurity area, such as AI models, zero-knowledge proofs (ZKPs), and quantum-safe payment systems, identifying how quantum computing could introduce new threats or enhance existing systems. It also suggests potential quantum-resistant solutions, such as post-quantum cryptography and hybrid encryption, while posing critical research questions that need to be addressed to ensure robust and scalable cybersecurity in the quantum computing field.

In Table 2, the research questions are systematically formulated by first identifying critical cybersecurity challenges posed by quantum computing (e.g., threats to RSA, ECC, and AES via Shor's and Grover's algorithms) and then pairing each question with proposed quantum-resistant solutions (e.g., post-quantum cryptography, QKD, or hybrid systems).

Moreover, we summarize key research problems and the proposed solutions suggested in the review paper on quantum computing and security. It looks like how quantum computers affect a variety of cryptographic systems, including those issues posed by Shor's and Grover's algorithms. In addition, solutions such as quantum key distribution (QKD) for secure information forward, post-quantum cryptography (PQC) for quantum cryptography, and hybrid systems to perform shift from classical to quantum systems. Addresses challenges in the implementation of quantum-resistant cryptography, the influence on blockchain and cloud security, and also the capacity of the quantum Internet to boost cybersecurity.

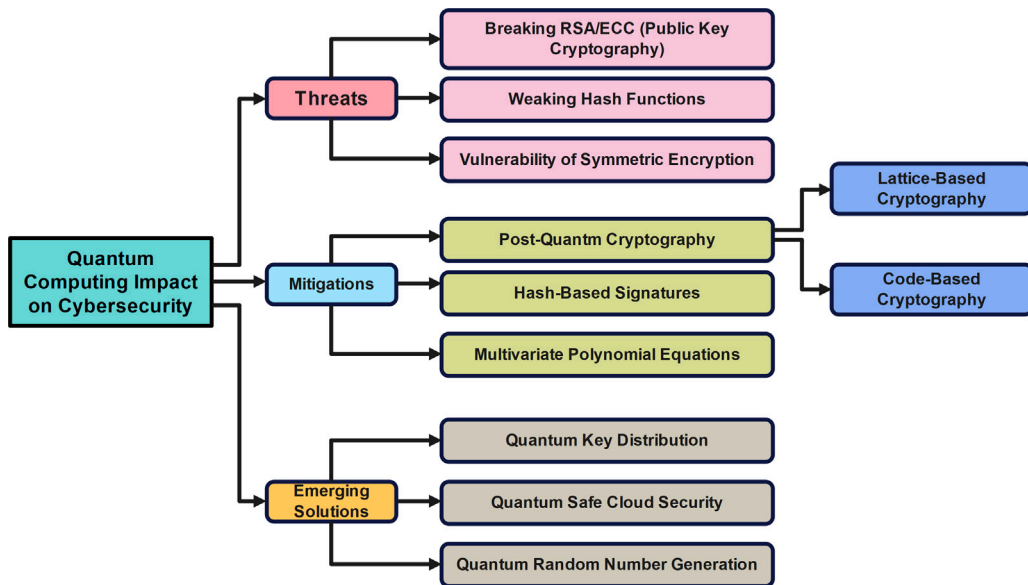


Fig. 1. Impact of quantum computing on cybersecurity: Threats, mitigations, and emerging solutions.

Table 1

Impact of quantum computing on cybersecurity: Risks, mitigations, and research questions.

Aspect	Risks/Opportunities	Mitigations	Research questions
Quantum-Safe AI	Faster quantum-trained models may increase adversarial threats.	Apply post-quantum cryptography (PQC) in AI pipelines.	How can quantum-safe AI standards prevent misuse?
Zero-knowledge proofs	Quantum attacks threaten ZKPs in privacy protocols.	Build ZKPs with PQC-resistant primitives.	Can we balance ZKP efficiency and quantum resistance?
Quantum access control	Speedups risk biometric matching integrity.	Use PQ algorithms for biometric templates and matching.	How to integrate quantum access with edge/cloud?
Secure MPC	MPC privacy may degrade with quantum algorithms.	Use quantum-resilient communication protocols.	What trade-offs exist between PQ security and performance?
Autonomous systems	Encrypted V2X/drone channels at risk from quantum decryption.	Adopt PQ encryption for real-time vehicle communication.	Can PQ crypto ensure low-latency secure operation?
Federated learning	Model aggregation may be attacked via quantum means.	Use homomorphic and PQ encryption in federated setups.	What makes PQ federated learning scalable and efficient?
Forensic tools	Quantum tools may compromise encrypted evidence.	Secure tools using PQ techniques to protect investigations.	How can forensics use quantum safely without new risks?
National frameworks	Quantum power in adversarial nations risks global crypto.	Promote global PQ cryptography standards and cooperation.	How to create ethical quantum governance frameworks?
Distributed quantum systems	Increased quantum power could aid offense/defense.	Use layered, hybrid cryptography for protection.	How to balance quantum benefits with cyber threats?
Payment systems	Digital payments vulnerable to quantum decryption.	Shift to quantum-safe payment protocols.	What are the trade-offs in PQ payment adoption?
Hardware security	Side-channel risks may grow with quantum tech.	Use PUFs and PQ-secure hardware designs.	How can PQ hardware support legacy systems?

1.1. Novelty of this review paper

This paper offers a unique interdisciplinary synthesis of quantum computing's impact on cybersecurity by critically evaluating cutting-edge quantum-resistant solutions like lattice-based cryptography, QKD, and hybrid systems, while addressing underexplored domains such as quantum-safe blockchain and IoT. Unlike prior surveys, it integrates theoretical advancements with practical deployment challenges, providing a holistic roadmap for transitioning to post-quantum security. Its originality lies in bridging gaps between cryptographic theory, hardware constraints, and policy frameworks, while proposing actionable strategies for global standardization and cross-sector collaboration in the quantum era.

Table 2

Questions and solutions addressed in the review paper.

Research questions	Proposed solutions
How will quantum computing impact existing cryptographic systems?	Quantum computers, using Shor's algorithm, can efficiently solve integer factorization and discrete logarithm problems, undermining the security of RSA and ECC.
What are the potential threats to symmetric encryption?	Grover's algorithm poses a threat by offering a quadratic speedup for brute-force attacks, reducing the effective security of symmetric encryption like AES.
How can we secure communication against quantum attacks?	Quantum Key Distribution (QKD), using protocols like BB84, provides secure key exchange leveraging quantum mechanics, ensuring that communication is immune to eavesdropping.
What cryptographic methods are quantum-resistant?	Post-Quantum Cryptography (PQC) methods, including lattice-based, multivariate-based, and code-based cryptosystems, are designed to be resistant to quantum attacks.
How can we bridge the gap between classical and quantum-resistant systems?	Hybrid cryptographic systems combining classical encryption with quantum-resistant algorithms offer a transition path while current standards evolve.
How can blockchain technologies be made quantum-safe?	Quantum-safe blockchain protocols are being developed by integrating quantum-resistant cryptographic algorithms to secure consensus mechanisms and data integrity against quantum attacks.
What are the challenges in implementing quantum-resistant cryptography?	Challenges include algorithm maturity, performance overhead, interoperability with existing systems, and scalability issues in real-world applications.
How will quantum computing affect cloud security?	Cloud providers are preparing for quantum risks by adopting hybrid cryptographic solutions and integrating post-quantum cryptographic standards for enhanced security.
What are the implications of quantum computing for IoT and edge security?	Quantum cryptographic methods are being tailored for IoT and edge computing environments where low power consumption and high security are critical, with quantum-resistant protocols ensuring confidentiality.
What is the future of quantum internet and its role in cybersecurity?	Research on quantum internet focuses on using quantum entanglement and teleportation for secure communication, which could fundamentally enhance data protection in the quantum era.

1.2. Our major contribution

This paper makes several significant contributions to the field of quantum computing and cybersecurity, addressing both theoretical and practical challenges. The key contributions are as follows:

- **Comprehensive Threat Analysis:** We provide a detailed examination of the vulnerabilities introduced by quantum computing to classical cryptographic systems, including RSA, ECC, and AES, through the lens of Shor's and Grover's algorithms. Our analysis highlights the exponential and quadratic speedups offered by these algorithms, respectively, and examines their implications for data confidentiality, integrity, and authentication.
- **Mitigation Strategies and Post-Quantum Cryptography (PQC):** We explore emerging quantum-resistant cryptographic techniques, such as lattice-based cryptography (e.g., Kyber, NTRU), code-based cryptography (e.g., McEliece), and hash-based signatures (e.g., XMSS). Our discussion includes their mathematical foundations, security guarantees, and practical implementation challenges.
- **Quantum Key Distribution (QKD) and Hybrid Systems:** We evaluate QKD as a secure key exchange mechanism leveraging quantum mechanics, emphasizing its resistance to eavesdropping. Additionally, we propose hybrid cryptographic systems that combine classical and quantum-resistant algorithms to facilitate a seamless transition to post-quantum security.
- **Application-Specific Solutions:** We address the impact of quantum computing on critical domains such as cloud security, blockchain, IoT, and edge computing. For each domain, we outline tailored quantum-resistant solutions, including post-quantum encryption for cloud data, quantum-safe blockchain protocols, and lightweight PQC algorithms for resource-constrained IoT devices.
- **Future Directions and Challenges:** We identify open research questions and practical hurdles in deploying quantum-resistant technologies, such as scalability, interoperability, and algorithmic maturity. Our recommendations emphasize the need for global standardization, performance optimization, and interdisciplinary collaboration to ensure robust cybersecurity in the quantum era.

- **Comparative Analysis:** We present a comparative analysis of traditional, modern, and quantum computing paradigms across key cybersecurity attributes (e.g., encryption strength, vulnerability to attacks, scalability), offering insights into their strengths and limitations.

By synthesizing the latest advancements in quantum algorithms and quantum-resistant cryptography, this paper serves as a foundational reference for researchers, practitioners, and policymakers working to safeguard digital infrastructures against quantum threats. Our work underscores the urgency of adopting quantum-safe solutions while paving the way for future innovations in the field.

1.3. Primary objectives of this review

Our review article investigates the effect of quantum computing on cybersecurity through the analysis of quantum algorithms' threats and the justification of defensive mechanisms and novel technological developments that maintain digital security. The main objectives of this paper are:

1. **Exploring the Quantum Threat Landscape:** We analyze how quantum computing can potentially maintain current cryptographic systems. In addition, we review the implications of quantum algorithms such as Shor and Grover algorithms on widely used encryption methods and protocols. The paper examines how quantum computing could undermine data privacy, authentication, and digital signature systems, highlighting the vulnerabilities that organizations and individuals face as quantum computing matures.

2. **Evaluation of mitigation and countermeasures:** We study post-quantum cryptography (PQC) because it develops cryptographic algorithms with quantum resistance to counter the growing threat posed by quantum computing. This article presents a summary of the current quantum-resistant algorithm standardization initiatives, including lattice-based cryptography, code-based cryptography, and multivariate polynomial systems. Additionally, this research analyzes the quantum key distribution (QKD), which is a secure communication approach that uses quantum mechanics to protect against quantum eavesdropping attacks.

3. **Review of emerging solutions and innovations:** In addition to PQC and QKD, this paper examines other emerging solutions such as hybrid cryptography, which combines classical and quantum-resistant algorithms to secure data in the transition period, and quantum-safe blockchain technologies that are being developed to withstand quantum adversaries. The review also looks at the potential for a quantum internet, which could offer secure communications based on quantum principles such as entanglement and quantum teleportation.

4. **Identifying Challenges and Future Research Directions:** Finally, the paper identifies the significant challenges in transitioning from classical to quantum-safe systems, including the scalability of quantum computing, the need for global standards in quantum cryptography, and the practical hurdles involved in deploying quantum technologies. We also discuss the need for continued innovation in both quantum-safe cryptography and quantum computing to ensure long-term security in an increasingly quantum-driven world.

Through this exploration, the paper aims to provide a comprehensive overview of the current state of research in quantum computing and cybersecurity, offering insight into the next generation of secure systems in the quantum era. The review highlights the urgent need for collaboration between quantum scientists, cryptographers, and cybersecurity experts to address the looming threats posed by quantum computing and develop the necessary tools for a secure digital infrastructure.

The Fig. 2, outlines the structured roadmap of a review paper on the intersection of quantum computing and cybersecurity. It begins with an introduction and preliminary background (1–2), followed by an exploration of quantum computing concepts and algorithms (3). The core focus is on the impact of quantum computing on cybersecurity (4), mitigations and proposed solutions (5), and emerging research (6). A comparative analysis between quantum computing and cybersecurity (7) is included, along with challenges and future directions (8), concluding with a summary (9). The roadmap provides a comprehensive framework for analyzing how quantum computing influences cybersecurity and ongoing efforts to address associated risks.

2. Preliminary

In this section, we provide explanations of the mathematical symbols and their descriptions used throughout the review. A detailed explanation is provided in Table 3.

3. Quantum computing: Concepts algorithms

3.1. Classical vs. Quantum computing

The core difference between classical and quantum computing lies in their fundamental processing principles rather than hardware constraints [20]. Classical computers use binary bits ($b \in \{0, 1\}$) and Boolean logic gates (AND, OR, NOT, XOR) for computation [21,22], often facing exponential complexity ($O(n!)$) in problems like the Traveling Salesman Problem (TSP) [23]. In contrast, quantum computing employs qubits ($|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$) that exploit superposition and entanglement (e.g., Bell states $|\psi_{entangled}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$), enabling parallel computation and outperforming classical methods in tasks like cryptography [24–26]. The interplay between classical and quantum computing highlights their coexistence, with classical systems providing foundational support for quantum advancements (e.g., circuit simulation) while quantum systems excel in specialized applications, collectively driving innovation in computing [27].

Roadmap of The Review Paper

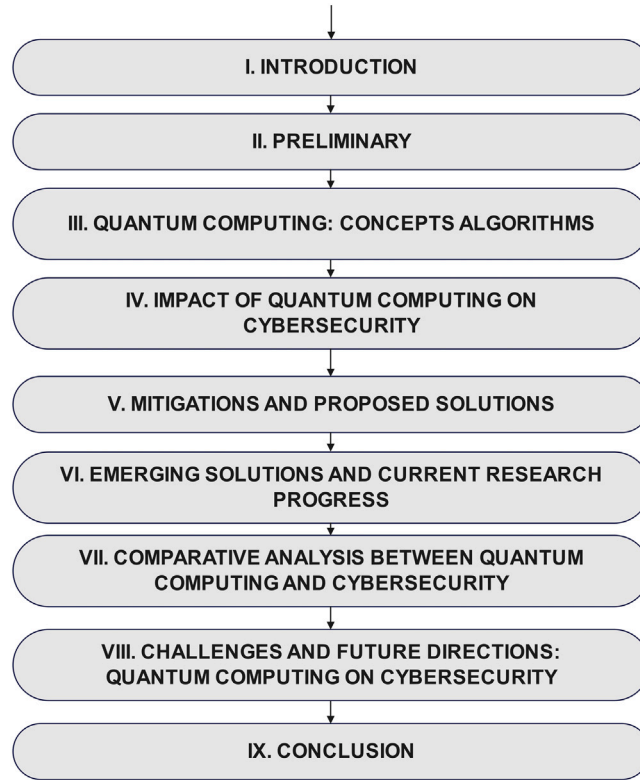


Fig. 2. Roadmap of the review article.

Table 3

Preliminary table of mathematical symbols and descriptions.

Symbol	Type	Description
b	Bit	Classical binary bit that can be in state 0 or 1
$ \psi\rangle$	Qubit	Quantum bit (qubit) in superposition state $\alpha 0\rangle + \beta 1\rangle$
α, β	Complex numbers	Probability amplitudes in qubit state, where $ \alpha ^2 + \beta ^2 = 1$
$ 0\rangle, 1\rangle$	Basis states	Orthogonal basis states of a qubit
$ \psi_{entangled}\rangle$	Entangled state	Bell state representing quantum entanglement $\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$
N	Integer	Composite number to be factored (in RSA) $N = p \cdot q$
p, q	Prime numbers	Large primes used in RSA key generation
e, d	Integers	Public and private exponents in RSA cryptosystem
$\varphi(n)$	Euler's totient	$\varphi(n) = (p-1)(q-1)$ for RSA modulus n
G	Point	Generator point on elliptic curve (for ECC)
k	Integer	Private key in ECC ($Q = k \cdot G$)
$H(M)$	Hash function	Cryptographic hash of message M
$\mathbf{A}, \mathbf{s}, \mathbf{e}$	Matrix/Vectors	Components in LWE problem: $\mathbf{A} \cdot \mathbf{s} + \mathbf{e} = \mathbf{b}$
$f_i(x_1, \dots, x_n)$	Polynomial	Multivariate quadratic polynomial in MQ cryptography
H_i	Hash function	Cryptographic hash function in hash-based signatures
S_i	Signature	Digital signature in XMSS scheme $S_i = H_i(\mathbf{m}, \mathbf{r}_i)$
K_{shared}	Key	Shared secret key in QKD protocols
$E_{classical}(m)$	Function	Classical encryption function (e.g., RSA, AES)
$E_{quantum}(m)$	Function	Quantum-resistant encryption function (e.g., Kyber, NTRU)
$\text{QFT}(x\rangle)$	Transform	Quantum Fourier Transform of state $ x\rangle$
$O(\cdot)$	Notation	Time complexity notation (e.g., $O(\log^3 N)$ for Shor's algorithm)

3.1.1. The key mathematical differences: Classical vs. Quantum

Information replication and management create the main mathematical differentiator between classical and quantum computing. The key points are explained below:

Superposition vs. Binary States: A classical computing system enables bits to exist as 0 or 1 during any computational period [28–31]. The vector states of quantum computers allow qubits to exist simultaneously in a superposition of the $|0\rangle$ and $|1\rangle$ states, according to research on quantum computing [32]. Using this capability, quantum computers process an extremely wide number of simultaneous possibilities. Superposition enables quantum computers to execute multiple possibilities in parallel due to their unique ability, which a classical computer lacks [27,33,34]. This behavior is demonstrated by an example in [35]. The parallel processing capabilities are the basis for which quantum computers reach a drastic exponential speedup for specific problem domains.

Quantum Interference: The probabilities of quantum states interfere with each other through quantum interference processes, resulting in constructive or destructive outcomes [36]. Quantum computers benefit from this phenomenon to increase their likelihood of correct answers while significantly reducing incorrect answers [25,37]. Thus, interference is mathematically characterized by the evolution of quantum states under unitary operations, where the amplitude of a state can be manipulated with quantum gates (i.e., rotations and permutations) [38]. This interference mechanism is especially useful as quantum algorithms evolve the quantum state toward optimal success, where the probability of the correct result is maximized [39–41].

Entanglement vs. Classical Correlation: Although in classical computing, bits are correlated, they are independent in their states [42]. In quantum computing, qubits may become entangled, and such a non-classical correlation does not exist among classical systems [43]. Entanglement exists as a mathematical construct that cannot be described independently of other variables and requires simultaneous measurement of one qubit to affect the other regardless of spatial distance [44]. The power of entanglement makes quantum algorithms possible because it enables the creation of highly parallelized quantum states that solve complex problems faster than traditional computers [45]. It is through entanglement that Shor's algorithm employs quantum parallelism to rapidly factor large integers, since classical computers would require impractical amounts of time [46].

3.1.2. Why quantum systems can break classical encryption?

Quantum computers obtain their exceptional computational speed through their ability to handle superposition and interference phenomena, among other processes [47]. The treatment involves breaking the RSA and ECC encryption protocols, despite their basis on number-theoretic problems that classical computers find difficult [48].

Shor algorithm and RSA: RSA encryption bases its functionality on the ability to factor large numbers into their prime factors, as described in [49]. RSA maintains its security value because factoring large composite numbers remains out of reach for classical computers, as noted in [50]. RSA encryption faces a breakthrough due to the Shor algorithm, created in 1994, that performs large factoring in polynomial time, as explained in [51]. The Shor algorithm uses factoring time to solve integers, whereas the classical factoring problem increases exponentially with the number of digits, leading to a computational running time of $O(\log^3 n)$, where n represents the factored number. The Shor algorithm solves the periodicity problem associated with factoring, which mathematically uses the quantum Fourier transform with an exponential speed-up over classical algorithms that cannot achieve polynomial time [46]. Such a significant speed-up, exponential in nature, poses a direct threat to RSA and other similar encryption schemes.

Grover Algorithm and Symmetric Encryption: The quantum algorithm known as Grover's algorithm [52] offers symmetric encryption algorithms, such as AES, a speedup that grows as the square root of its key length. A classical brute-force character search takes $O(N)$ time, but Grover's algorithm finishes a search of an unsorted N -item database in $O(\sqrt{N})$ time using quantum computing. When using symmetric key encryption, the effective security strength of the key is reduced to half its original value. An AES-256 encryption would require 2^{256} operations using traditional computation but would require only 2^{128} operations if a quantum computer used Grover's or analogous cryptographic methods. Although not as severe as the exponential vulnerability of public key cryptography against Shor's algorithm, such a quadratic loss of security is still a significant decrease in symmetric encryption standards today [53].

The information processing mechanisms used in quantum computing differ fundamentally from those in classical computing [54]. Quantum computers operate using quantum mechanical principles by allowing qubits to remain in superposition and establishing entangled connections that enable interference fundamentally different from classical computing [4]. Quantum computers provide rapid performance on particular problems, although they currently support only a limited number of solvable problems, such as unstructured searches and integer factorization, which threaten current cryptographic solutions. The Shor and Grover quantum algorithms can break RSA and AES encryption, highlighting the need to develop quantum-proof cryptographic approaches due to this vulnerability [55].

In addition, in Table 4, we provide a detailed comparison between classical and quantum computing across various parameters, highlighting their fundamental differences. Classical computing operates on deterministic bits and relies on sequential logic, while quantum computing utilizes qubits, enabling superposition and entanglement for parallel processing. The discussion centers on quantum computing's advantage of significantly accelerated problem-solving capabilities, which result from using Shor's and Grover's algorithms. A comprehensive assessment of hardware demands, together with error correction methods is provided, along with an overview of vulnerabilities that quantum systems pose to classical encryption methods like RSA and AES, as presented in the table. The analysis includes a discussion of scalability difficulties, noise levels, and gate fidelity problems that affect quantum systems.

Table 4
Comparison between classical and quantum computing.

Parameter	Classical computing	Quantum computing
Basic unit	Bit (0 or 1)	Qubit (superposition of 0 and 1, can exist in multiple states)
State representation	Deterministic states (0 or 1)	Probabilistic states, superposition (e.g., $\alpha 0\rangle + \beta 1\rangle$)
Computation model	Sequential execution of logical operations	Parallel computation via superposition and entanglement
Information processing	Processes one value at a time (serially)	Can process multiple values simultaneously (quantum parallelism)
Computational power	Limited by classical bit operations, exponential time complexity for certain problems (e.g., factoring large numbers)	Exponential speedup for certain problems (e.g., Shor's and Grover's algorithms)
Impact on cryptography	Relies on complexity of problems like integer factorization and discrete logarithms for security (e.g., RSA)	Can break current cryptographic systems like RSA and AES via Shor's and Grover's algorithms
Key algorithms	RSA, AES, DES, etc., based on classical algorithms like factorization and discrete logarithms	Shor's algorithm (integer factorization), Grover's algorithm (search problems)
Cryptographic vulnerability	Vulnerable to brute-force and advanced classical attacks (e.g., factorization of large numbers)	Vulnerable to Shor's algorithm (breaking RSA) and Grover's algorithm (reducing AES security)
Computational efficiency	Polynomial time complexity for many problems, but exponential time for others (e.g., NP-hard problems)	Polynomial or exponential speedup for specific problems (e.g., $O(n^3)$ for Shor's algorithm)
Gate operations	Logical gates (AND, OR, NOT)	Quantum gates (Hadamard, CNOT, T, Phase, etc.) which operate on qubits, enabling entanglement
Simultaneous computation	Serial computation (one operation after another)	Quantum parallelism allows multiple computations at once
Scalability	Limited by transistor size and quantum of hardware available	Scalability is a challenge (e.g., qubit coherence, gate fidelity) but theoretical exponential speedup for large quantum systems
Error correction	Classical error correction methods (e.g., Hamming, Reed-Solomon)	Quantum error correction codes (e.g., Shor's code, surface codes), but they require additional qubits and complexity
Hardware requirements	Classical processors (e.g., CPUs, GPUs, ASICs)	Quantum processors (e.g., superconducting qubits, trapped ions, topological qubits)
Noise	Decoherence	High noise and decoherence rates in current quantum hardware, limiting gate fidelity and coherence time
Gate fidelity	High fidelity, close to 100% for classical gates	Lower fidelity, quantum gates often suffer from errors and imperfections
Speedup potential	Speedup is typically limited (exponential time for certain problems)	Potential for exponential speedup (e.g., Shor's for integer factorization)
Memory	Memory is stored in bits, limited by hardware size	Memory is stored in qubits, and quantum systems can encode information in quantum states
Parallelism	No true parallelism in classical bits (aside from multi-core processors)	True quantum parallelism due to superposition and entanglement

3.2. Key quantum algorithms

3.2.1. Introduction to quantum algorithms

The initial quantum computing method for problem solving presents an entirely unique approach [56–58]. The main difference between classical computers and quantum computers lies in the way they handle information: classical machines operate in binary (0 or 1) states, while quantum computers process quantum bits, also known as qubits, which exist simultaneously in both 0 and 1 states [59]. This fundamental distinction allows quantum computation to enable algorithms to achieve exponential or quadratic speedup on various problems through parallel execution and interference when solving specific classes of computational tasks [60]. Quantum algorithms include two of the most influential methods in the form of the Shor algorithm and the Grover algorithm [61–63]. The significance of these algorithms lies in their ability to compromise encryption systems like RSA and AES [53].

3.2.2. Shor algorithm for integer factorization

The researcher Peter Shor presented his quantum algorithm to solve the integer factorization problem in 1994, known as Shor's algorithm [46]. The security of RSA encryption relies on a hard mathematical problem [64–66], but Shor's algorithm, running on quantum computers, can efficiently decrypt numbers even when their keys are extremely large [67,68].

The Integer Factorization Problem: Given a composite integer N , the goal is to find its prime factors such that:

$$N = p \cdot q \quad (1)$$

The statement contains two prime numbers designated by p and q . The integer factorization methods, such as trial division, Pollard's rho algorithm, and the quadratic sieve, require exponential running time to solve problems in their worst cases. Through Shor's algorithm, large numbers become easier to factorize, and the process remains fast.

Classical Approach to Factoring: The classic way to solve the integer factorization problem is by attempting to divide N by all numbers below N (or by using a heuristic search). The property that makes factorization difficult is that the number of possible divisors increases exponentially with N . The ease with which we cannot factor these numbers is the basis of RSA security, and typical key sizes for RSA keys are on the order of 2048 to 4096 bits. The best classical algorithms still require time proportional to $O(\exp(n^{1/3}))$ even in these cases, where n is the number of bits of N .

Quantum Speedup in Shor's Algorithm: The previous statement refers to the prime numbers p and q . The integer factorization methods known as trial division, along with Pollard's rho algorithm and the quadratic sieve, require exponential running time in worst-case scenarios. Shor's algorithm can solve factorization problems quickly despite the difficulty posed by large numbers:

$$a^r \equiv 1 \pmod{N} \quad (2)$$

The value r serves as a critical point in the factorization process because obtaining it enables classical methods to find N 's factors within polynomial runtime. The quantum component of Shor's algorithm efficiently determines r .

Quantum Fourier Transform (QFT): in Shor's algorithm [20] enables efficient period-finding by transforming quantum states as $\text{QFT}(|x\rangle) = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} e^{2\pi i \frac{kx}{n}} |k\rangle$, where n is the system dimension and x a quantum state [21,22]; this quantum parallelism allows Shor's algorithm to factor integers N by first finding the order r of a random a ($1 < a < N$) through quantum computation, then classically deriving factors via $p = \gcd(a^{r/2} - 1, N)$ and $q = \gcd(a^{r/2} + 1, N)$ when r is even and $a^{r/2} \not\equiv -1 \pmod{N}$ [23], achieving polynomial complexity $O((\log N)^3)$ compared to classical exponential-time factorization [24–26], demonstrating quantum advantage in cryptography [27].

Implications for Cryptography: The ability of Shor's algorithm to factor numbers in polynomial time endangers RSA security, as well as other cryptographic schemes that rely on hard prime factorization [69]. A quantum computer can break a 2048-bit RSA key within a reasonable time frame according to [70]. The cryptographic community must develop quantum-resistant and post-quantum cryptography for this compelling reason.

3.2.3. Grover algorithm for unstructured search

Through Shor's algorithm, factorization of integers is possible [71], and Grover's algorithm achieves quadratic speedup for unstructured search problems [72]. These algorithms deliver important benefits to symmetric key encryption systems such as AES because AES security relies on the difficulty of performing complete key space searches [73].

The Unstructured Search Problem: The unstructured search problem involves locating an index x within an unsorted N -element database using a search function f where $f(x) = 1$. Traditional computer systems require a search process that examines all elements of the database, resulting in a worst-case time complexity of $O(N)$. The brute-force attack exploits this problem in cryptographic algorithms by requiring the complete checking of all potential key combinations.

Grover's Search Algorithm: solves unstructured search with $O(\sqrt{N})$ complexity, offering quadratic speedup over classical brute-force methods [74]. It begins with a uniform superposition $|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$ and iteratively applies: (1) an oracle $O|x\rangle = (-1)^{f(x)}|x\rangle$ that phase-inverts the target state $|x^*\rangle$ [75], and (2) a diffusion operator that amplifies this marked state through amplitude reflection [76]. After approximately $\frac{\pi}{4}\sqrt{N}$ iterations, the correct solution is measured with high probability, demonstrating quantum advantage for search problems that would require $O(N)$ classical evaluations.

Implications for Symmetric Key Cryptography: The key security provided by symmetric key cryptography is effectively reduced by a factor of 2 according to the research presented in [77]. AES-256 benefits from Grover's algorithm by requiring only 2^{128} operations instead of the original 2^{256} operations classically [8]. The security level of modern algorithms is weakened post-Grover algorithm despite the encryption method itself remaining intact [78]. Post-quantum cryptography designers may need longer key lengths because Grover's algorithm provides quantum speed-up, effectively reducing the complexity of brute-force attacks [79].

Summary of Quantum Speedup for Search: The essential contribution of Grover's algorithm is providing unstructured search problems with a speed-up that reaches a quadratic level [80–82]. Identifying this speedup becomes crucial because it targets threats frequently encountered during cryptographic key searches [83]. Grover's algorithm weakens symmetric encryption by reducing the number of operations required, though it impacts public-key cryptography less than Shor's algorithm [84]. Shor's algorithm and Grover's algorithm represent the two biggest discoveries in quantum computation, having a significant impact on the field of cryptography [85].

Grover's algorithm achieves quadratic speedup for unstructured search problems and can potentially break some symmetric keys like AES [86], while Shor's algorithm can efficiently factor large numbers and break public-key encryption schemes like RSA [87]. These quantum algorithms demonstrate the power of quantum computers, making quantum-resistant cryptographic

Table 5
Key quantum algorithms and their impact on cryptography.

Quantum algorithm	Classical counterpart	Impact on cryptography
Shor's algorithm [89]	Classical integer factorization algorithms	Breaks RSA, DSA, and DH by factoring large integers efficiently, rendering these schemes insecure against quantum attacks.
Grover's algorithm [90]	Classical brute-force search	Halves the effective key length of symmetric encryption (e.g., AES), requiring longer keys to maintain security.
Quantum Fourier Transform (QFT) [91]	Classical Fourier transform	Enables period finding, fundamental to Shor's algorithm; indirectly threatens cryptographic systems based on periodicity.
Quantum Phase Estimation (QPE) [92]	Classical estimation algorithms	Aids algorithms like Shor's and may impact schemes based on eigenvalue or modular arithmetic problems.
Quantum simulations [93]	Classical simulation algorithms	Could improve cryptanalysis and aid in evaluating quantum-resistant cryptosystems.
Quantum random walks [94]	Classical random walks	Speeds up search tasks and may weaken cryptographic schemes based on random processes or hash functions.
Quantum Key Distribution (QKD) [95]	Classical key exchange algorithms	Offers secure key exchange immune to quantum attacks, potentially replacing RSA/DH for key distribution.

protocols necessary. Therefore, studying and establishing post-quantum cryptography is “uniquely important” to ensure the future security of digital systems enabled by quantum technologies [88].

Table 5 presents simplified explanations about the basics of quantum algorithms and their classical comparisons along with their impacts on cryptography. The quadratic advantage of Grover's algorithm threatens symmetric encryption including AES, while Shor's algorithm can quickly break RSA and DH public-key systems, which rely on integer factorization for security. The necessary fundamental components for solving quantum periodicity and breaking eigenvalue problems include the quantum Fourier transform (QFT) together with quantum phase estimation (QPE). The table also explains quantum key distribution (QKD) as a prototype for tamper-evident key exchange, based on quantum mechanics principles. Quantum algorithms serve a double purpose since they both push current cryptography standards toward obsolescence and drive the development of future secure systems toward next-generation security.

3.3. Quantum hardware and software

Quantum computing has established an innovative processing method based on quantum mechanical phenomena such as superposition, entanglement, and quantum interference, enabling it to operate beyond classical computing theory [1]. The process of building functional quantum computers depends on both the development of dependable quantum hardware and efficient software systems [96]. Quantum hardware advancement has become more prominent since scientists developed superconducting qubits, as well as trapped ions and topological qubits, in parallel [97]. The successful deployment of these systems faces several obstacles in terms of scalability, coherence time, error rate reduction, and maintaining the reliability of quantum software on such devices [98].

3.3.1. Quantum hardware technologies

The realization of quantum bits (qubits) serves as the basic physical component of quantum hardware units [56]. Due to their ability to implement quantum superposition of states, quantum computers can manage significantly more information than conventional systems can [99]. Multidimensional quantum hardware systems have different physical qubit representations that yield distinct strengths and weaknesses with respect to operation time, accuracy, reliability, and scalability across various platforms (such as in [100]). This research centers on two extensively studied technologies by performing loop-graph-based congestion time evaluations for superconducting qubits and trapped ions.

Superconducting bits: One widely developed quantum hardware platform is superconducting qubits [101]. The qubits in these systems are Josephson junctions whose quantum mechanical behavior occurs at macroscopic scales. Using standard lithographic techniques, superconducting qubits are manufactured in a way that is compatible with semiconductor fabrication processes and thus are potentially scalable [102,103]. The mathematical description of a superconducting qubit takes the form of a nonlinear LC circuit, whose quantum properties result from quantum tunneling and superposition enabled by the junction. The superconducting qubit has its Hamiltonian described as:

$$H = 4E_C (\hat{n} - n_g)^2 - E_J \cos(\hat{\phi}) \quad (3)$$

where: E_C is the junction charging energy, E_J is the Josephson energy, $\hat{n}$ is the number operator (representing charge), n_g is the external charge bias, $\hat{\phi}$ is the phase difference across the junction.

The system operates with discrete energy levels, while the qubit exists between the states $|0\rangle$ and $|1\rangle$, which are controlled by microwave pulses applied to the system. These superconducting qubits use energy eigenstates as representations of their qubit states through the Cooper-pair tunneling mechanism. The major challenges with superconducting qubits lie in improving decoherence times and gate fidelities [104,105]. Decoherence is the primary limiting factor in preserving quantum information, as interactions with the environment lead to information loss [106]. The duration of quantum coherence (T_2) is reduced when the qubits interact with their surrounding environment, thus decreasing the number of applicable quantum gate operations.

Trapped-Ion Qubits: The fundamental building blocks of trapped-ion quantum computing consist of isolated ions confined by electromagnetic fields and controlled using lasers, according to the literature [107]. The internal energy states of the ions function as qubits, while the external motional states of the ions enable both entanglement and qubit interaction, according to Lu and Wang [108]. The mathematical description of a trapped-ion qubit typically involves the Hamiltonian for a single ion in a potential well, which couples the internal electronic state with the motional state of the ion:

$$H = H_{\text{internal}} + H_{\text{motion}} + H_{\text{interaction}} \quad (4)$$

where:

$H_{\text{internal}} = \hbar\omega_0 (|0\rangle\langle 0| - |1\rangle\langle 1|)$ represents the energy of the internal states of the ion.

$H_{\text{motion}} = \hbar\omega_0 \hat{n}$ represents the harmonic motion of the ion, where $\hat{n}$ is the number operator for the motion modes.

$H_{\text{interaction}} = \hbar\Omega (|0\rangle\langle 1| + |1\rangle\langle 0|)$ represents the interaction between the internal and motional states.

Trapped-ion qubits are known for their long coherence times, often in the range of several seconds [109], making them highly suitable for precision-based quantum algorithms. Furthermore, quantum gates in ion traps are implemented using laser pulses that induce transitions between internal qubit states, and entanglement is generated through laser-mediated interactions between ions. The key challenge for trapped-ion systems is the difficulty in scaling to a large number of qubits [110]. Although a small-scale trapped-ion system can achieve high gate fidelities and long coherence times, it becomes technically demanding to arrange and control large arrays of ions. Moreover, quantum operations in ion traps are typically local, which means that creating large-scale entanglement throughout the system requires efficient and fast communication between qubits [111].

Topological Qubits (Emerging Technology): Topological qubits, which are based on anyonic particles that exist in specific two-dimensional materials, represent a promising emerging quantum hardware platform [112]. These qubits are based on the topological properties of these particles, which means that they are inherently resistant to local perturbations, offering the potential for fault tolerance [113]. The topological qubits are mathematically described by the topological ground state of a system of anyons, particles that obey braid statistics [114]. Anyonic excitations and their states are inherently protected from decoherence [115], and the braiding of anyons defines the quantum state of a topological qubit. However, topological qubits are still in the experimental stage [116], and major challenges remain for their physical realization.

3.3.2. Scalability and reliability challenges in quantum hardware

As we explore the development of quantum hardware, the issues of scalability and reliability remain two of the most significant hurdles to creating practical quantum computers. Quantum systems face intrinsic challenges such as decoherence, error correction, and control precision. Overcoming these obstacles is essential for the realization of scalable and dependable quantum computing systems.

Decoherence and Error Rates: The loss of quantum information to environmental interactions, called decoherence [117], is a primary challenge for quantum hardware. Different qubit technologies exhibit significantly different decoherence times, with superconducting qubits generally having shorter coherence times than trapped ions [118]. To mitigate decoherence's detrimental effects, quantum error correction schemes have been developed that encode quantum information across multiple physical qubits, such as the Shor code and surface codes [119,120]. However, these approaches require substantial overhead, and their practical implementation demands a large number of physical qubits. The decoherence rate γ can be mathematically described using the Lindblad equation, which models how a quantum system's density matrix evolves under environmental interactions:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \} \right) \quad (5)$$

where ρ is the density matrix of the quantum system, H is the system's Hamiltonian, and L_j are the Lindblad operators that describe system–environment interactions.

Gate Fidelity and Control Precision: Quantum gate fidelity measures how closely an implemented quantum gate operation matches the desired operation [121]. The physical implementation of gates in superconducting qubits and trapped ion systems is challenging, relying on complex control protocols; thus, achieving high gate fidelity is crucial for successful quantum algorithm execution [109,110]. The fidelity F of a gate is defined as:

$$F = \left| \langle \psi_{\text{target}} | U_{\text{actual}} | \psi_{\text{input}} \rangle \right|^2 \quad (6)$$

where U_{actual} represents the implemented gate operation, and ψ_{input} and ψ_{target} denote the initial and target quantum states, respectively. High gate fidelity requires minimizing errors from control signal imperfections and reducing environmental noise effects.

Table 6
Comparison of quantum hardware and software.

Aspect	Classical hardware/software	Quantum hardware/software
Hardware technology	Semiconductor processors (e.g., CPUs, GPUs, ASICs)	Superconducting qubits, trapped ions, photonics, topological qubits
Qubit representation	N/A	Qubits represented by physical systems (e.g., superposition of quantum states)
Scalability	Scalable due to advancements in semiconductor technology	Limited scalability due to qubit decoherence, gate fidelity, and error rates
Error correction	Classical error correction schemes (e.g., Hamming, Reed-Solomon)	Quantum error correction codes (e.g., Shor's code, surface codes) for improving qubit reliability
Quantum software platforms	Software such as C/C++, Python, Java for classical programming	Quantum software platforms (e.g., Qiskit, Cirq, Quipper) for quantum circuit design, simulation, and execution
Programming languages	High-level languages like Python, C, Java	Quantum programming languages (e.g., Qiskit, Cirq, Quipper) designed to express quantum algorithms
Gate operations	Logic gates (AND, OR, NOT)	Quantum gates (Hadamard, CNOT, T-gates) that manipulate qubits in quantum states
Quantum simulation	Simulations based on classical hardware (classical computational models)	Quantum simulators and quantum circuits using actual or simulated quantum hardware
Control precision	High precision in classical systems	High precision required for qubit manipulation, challenging with current quantum hardware
Reliability	High reliability with classical components	Current quantum hardware suffers from noise, decoherence, and imperfections in qubit manipulation
Development stage	Fully mature, with widespread application across industries	In early stages, with significant research and development focused on hardware and software reliability and scalability

3.3.3. Quantum software: Progress toward scalability

Unlike quantum hardware, quantum software plays a crucial role in building practical quantum computers [122]. Moreover, since quantum software must efficiently interface with the underlying quantum hardware, execute quantum algorithms, and implement quantum error correction and optimization [123], it requires design choices and decisions beyond those needed in classical computing. To help programmers write quantum algorithms more easily (such as in Example Listing 2), quantum programming languages including Qiskit, Cirq, and Quipper have been developed [124]. Quantum software frameworks enable developers to create algorithms using quantum gates that represent unitary matrices [125]. The single-qubit QFT implementation through software depends on Hadamard gates and controlled phase gates to perform a sequence of operations, as described by this matrix:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (7)$$

These frameworks manage the compilation of quantum circuits, ensuring they are properly mapped to specific hardware architectures while optimizing gate fidelity and minimizing quantum resource requirements. Recent advancements in quantum hardware and software are progressing rapidly, which breakthroughs with superconducting qubits and trapped ions [39]. While error rates are decreasing and coherence times are improving, the main barrier to achieving large-scale qubit systems remains a significant challenge [126]. The development of quantum software aims to meet quantum hardware requirements while serving as the foundation for creating and optimizing quantum algorithms [127]. Over the next ten years, additional hardware and software advancements will be required, along with fault-tolerant quantum architecture design and error-correcting code development, to enable the production of reliable, scalable quantum computers.

Table 6 compares the key characteristics of classical and quantum hardware and software, highlighting their differences and challenges. Classical systems are based on mature semiconductor technology and conventional programming languages that offer high reliability and scalability. In contrast, quantum systems utilize emerging technologies such as superconducting qubits and trapped ions, which must address challenges like qubit decoherence, noise, and error correction. Quantum software platforms such as Qiskit and Cirq enable the creation and simulation of quantum circuits, though they remain under development. This comparison underscores the but promising potential of quantum systems to revolutionize computation, despite significant hurdles in reliability, precision, and scalability.

4. Impact of quantum computing on cybersecurity

4.1. Quantum threats to cryptography

The advent of quantum computing will enable computational capabilities that can compromise multiple currently-used cryptographic protocols [128,129]. The ability of quantum computers to solve specific computational problems poses a significant threat,

Table 7

Quantum impact on cryptographic systems and mitigations.

System	Quantum threat	Impact	Mitigation
RSA	Shor's algorithm	Breaks integer factorization.	Lattice-, hash-, or code-based crypto.
ECC	Shor's algorithm	Solves elliptic curve DLP.	Post-quantum ECC or lattice-based.
DSA	Shor's algorithm	Solves discrete log problem.	Lattice- or hash-based signatures.
AES-128	Grover's algorithm	Cuts brute-force to $O(2^{64})$.	Use AES-256 or higher.
AES-256	Grover's algorithm	Cuts to $O(2^{128})$.	AES-512 or PQ symmetric crypto.
SHA-256	None	Hash remains secure.	Continue use in PQ schemes.
SHA-3	None	Stronger hash, quantum-safe.	Use in future systems.
Hash-based sigs	None	Quantum-resistant.	XMSS, SPHINCS+.
Lattice-based	None	Secure vs. quantum attacks.	Kyber (encryption), FALCON (signing).
Code-based	None	Quantum-safe (e.g., McEliece).	Public-key encryption.
Multivariate crypto	None	Resistant to Shor's.	Rainbow for signing.
QKD	None	Physical-layer secure key exchange.	Combine with classical crypto.
RSA-CRT	Shor's algorithm	Breaks via factoring.	Use PQ alternatives.
ElGamal	Shor's algorithm	Solves DLP.	Switch to PQ crypto.
Blowfish	Grover's algorithm	Reduces key space.	Upgrade to AES-256.
Twofish	Grover's algorithm	Brute-force at $O(2^{128})$.	Use larger keys or AES-512.
Lamport Sig.	None	One-time, hash-based.	Winternitz or SPHINCS+.
Merkle trees	None	Safe if hashes are PQ.	PQ blockchain structures.
PQ signatures	None	Built for quantum safety.	XMSS, SPHINCS+, Lizard.
PQ blockchain	None	Current chains vulnerable.	Use lattice/hash-based crypto.

as these machines can perform operations infeasible for classical computers. Most modern encryption systems, including public-key and symmetric-key cryptography, rely on these fundamental computational problems to secure electronic banking and e-commerce applications [130]. The following section examines how quantum algorithms threaten the cryptographic field by exploiting these mathematical problems.

Current secure communication networks predominantly implement public-key cryptosystems such as RSA, ECC, and DSA [131–133]. These systems rely on mathematical problems believed to be intractable for classical computers [134]. However, quantum computers can efficiently solve these problems using Shor's algorithm, thereby breaking the security of these cryptographic systems.

The Table 7, details how quantum computing affects specific cryptographic systems and their corresponding protection solutions. The report explains how public-key cryptosystems RSA, ECC, and DSA are vulnerable to Shor's algorithm, requiring migration to post-quantum algorithms such as lattice and hash-based systems. Furthermore, Grover's algorithm necessitates increased key sizes for AES and Blowfish to maintain security standards, as it reduces the time efficiency of brute-force attacks against these symmetric encryption schemes. Most experts identify lattice-based [135–137], code-based [138,139], and multivariate cryptography [140–142], as leading to quantum-resistant solutions, while hash-based systems like SHA-256 and SHA-3 are expected to remain secure in the post-quantum era. The analysis table demonstrates the importance of Quantum Key Distribution (QKD) for secure key exchange as well as quantum-resistant blockchain protocols for maintaining data integrity and security against quantum threats.

4.2. Risks to digital signatures and authentication

Digital signatures, PKI, and authentication protocols face significant quantum threats [143,144], as Shor's algorithm breaks RSA ($O((\log N)^3)$ factorization [145]), ECC (solving ECDLP [146]), and DSA (solving DLP [147]). RSA's security relies on factoring $n = p \times q$ where private key d satisfies $e \cdot d \equiv 1 \pmod{\varphi(n)}$, while ECC uses scalar multiplication $Q = kP$ and DSA depends on $y = g^x \pmod{p}$ - all vulnerable to quantum attacks [148]. The Phased Rekeying Protocol offers quantum-resistant security by mitigating these vulnerabilities while maintaining authentication and non-repudiation properties [149,150].

4.3. Risks to Public Key Infrastructure (PKI)

Public Key Infrastructure (PKI) relies on the security of private-public key pairs for authentication, digital signatures, and key exchange [151,152], but quantum computers threaten this model by using Shor's algorithm to derive private keys from public certificates [153]. This vulnerability would compromise Certificate Authorities (CAs) by enabling forged digital certificates and undermine authentication protocols like TLS/SSL [154], allowing quantum adversaries to decrypt sessions and impersonate servers [155], effectively collapsing the PKI trust framework.

4.4. Risks to authentication protocols

Authentication protocols like Kerberos, OAuth, and SAML rely on vulnerable public-key cryptography (RSA and ECC) for key exchange and authentication [156]. Kerberos uses RSA for initial key exchange, exposing it to quantum attacks that could steal private keys [157]. OAuth's token-based security is threatened as quantum computers could forge RSA/ECC-signed tokens [158]. Similarly, SAML's SSO infrastructure depends on quantum-vulnerable digital signatures for identity provider authentication [159], making all three protocols susceptible to quantum cryptanalysis.

4.5. Preventive measures

Current cryptographic schemes lack quantum resistance, threatening digital signatures and authentication systems. Migration to post-quantum algorithms is essential, including: (1) lattice-based Kyber (encryption) and FALCON (signatures), (2) quantum-resistant digital signatures like XMSS and SPHINCS+, and (3) the coding theory-based McEliece cryptosystem. While Quantum Key Distribution (QKD) can enable quantum-safe key exchange [160], comprehensive protection requires replacing vulnerable RSA/ECC/DSA schemes [161] with post-quantum cryptographic solutions across PKI systems, authentication protocols, and private-key algorithms to maintain security in the quantum era.

4.6. Data privacy and integrity risks in the quantum computing era

Quantum computing threatens current cryptographic systems that protect data privacy and integrity through symmetric and asymmetric encryption and hashing [162]. While classical methods like RSA (factorization), ECC (ECDLP), and ElGamal (DLP) have secured communications (e.g., TLS/SSL) and data storage [163], they become vulnerable to quantum attacks that can solve these previously-intractable problems efficiently. This exposes critical risks for both real-time data transmission and long-term stored information protected by these schemes.

RSA encryption $C = M^e \pmod n$ relies on the hardness of factoring $n = p \cdot q$, with decryption using private key d where $e \cdot d \equiv 1 \pmod{\phi(n)}$. While classically secure, Shor's algorithm enables quantum computers to factor n in $O((\log N)^3)$ time by: (1) finding prime factors p and q , then (2) computing d to decrypt C , completely breaking RSA's confidentiality [164].

ECC provides strong encryption with smaller keys than RSA, where a public key $Q = k \cdot G$ is derived from private key k and generator point G [165–167]. While secure classically, Shor's algorithm solves the ECDLP in $O((\log n)^3)$ time, enabling quantum computers to: (1) recover k from Q , and (2) decrypt any message M encrypted with Q . This breaks ECC's security in the quantum era.

The Advanced Encryption Standard (AES) employs symmetric-key cryptography [168], where classical brute-force attacks require 2^n operations for an n -bit key (e.g., 2^{128} for AES-128). Grover's quantum algorithm provides a quadratic speedup, reducing complexity to $O(2^{n/2})$:

- AES-128 security diminishes to $\sim 2^{64}$ operations, potentially vulnerable to future quantum systems
- AES-256 maintains stronger protection ($\sim 2^{128}$ operations), though reduced from classical estimates

Cryptographic hashes (e.g., SHA-256) provide fixed-length outputs [169] with:

- Pre-image resistance: Hard to reverse input from hash
- Collision resistance: Hard to find matching inputs

Grover's algorithm weakens security, reducing search complexity from $O(2^n)$ to $O(2^{n/2})$ (e.g., SHA-256 from 2^{256} to 2^{128}).

Digital signatures [170] become vulnerable as quantum computers break RSA, ECC, and DSA via Shor's algorithm, enabling forgery.

Quantum computing threatens long-term security [171], as harvested encrypted data [172] may be decrypted later, making forward secrecy critical [173].

Urgent adoption of quantum-resistant methods is essential to protect against Shor's and Grover's algorithms and ensure future data security [174].

5. Mitigations and proposed solutions

Post-quantum cryptography (PQC) encompasses cryptographic algorithms designed to resist attacks from quantum computers [14]. Shor's algorithm enables quantum computers to efficiently solve problems such as integer factorization, discrete logarithms, and elliptic curve discrete logarithms [175]. The goal of PQC is to develop encryption and authentication systems that remain secure against quantum adversaries. PQC techniques are primarily based on mathematical problems believed to be quantum-resistant, including lattice-based cryptography, code-based cryptography, multivariate quadratic (MQ) equations, and hash-based signatures.

In Table 8, we explain recent developments in the relevant topic, highlighting key research contributions and insights.

5.1. Lattice-based cryptography: A pillar of post-quantum security

Lattice-based cryptography relies on the computational hardness of the Shortest Vector Problem (SVP) and the Learning With Errors (LWE) problem [184]. These problems are currently considered resistant to quantum algorithms. The LWE problem involves determining a vector $s \in \mathbb{Z}_q^n$ such that $A \cdot s \approx b \pmod q$, given a matrix $A \in \mathbb{Z}_q^{m \times n}$ and a vector $b \in \mathbb{Z}_q^m$. The intractability of LWE forms the basis for its cryptographic security, as no efficient quantum algorithm exists to solve it in polynomial time.

Ring-based variants such as Ring-LWE, which operate over polynomial rings, are widely used in key exchange and digital signature schemes due to their efficiency [185]. The security of LWE stems from the difficulty of solving noisy linear systems, with the hardness closely tied to worst-case lattice problems [186]. The carefully chosen noise distribution ensures that distinguishing LWE samples from random remains challenging for both classical and quantum computers, preserving security against quantum attacks.

Table 8

Contribution made by the researchers and the novelty of the research.

Ref	Focused area	Contribution made by researchers
[176]	Navigating quantum security risks in networked environments: A comprehensive study of quantum-safe network protocols	This research analyzes quantum-induced vulnerabilities in key network protocols across the TCP/IP stack and evaluates post-quantum cryptographic solutions. It offers practical insights to guide secure protocol design in the quantum era.
[177]	From portfolio optimization to quantum blockchain and security: a systematic review of quantum computing in finance	This research reviews the intersection of quantum computing and finance, focusing on applications like portfolio optimization, fraud detection, and blockchain. It highlights key quantum algorithms and identifies future research directions for financial technologies.
[178]	Cyber diplomacy: defining the opportunities for cybersecurity and risks from Artificial Intelligence, IoT, Blockchains, and Quantum Computing	This research explores how emerging technologies—AI, IoT, blockchain, and quantum computing—shape cyber diplomacy, highlighting both opportunities and threats. It underscores the need for international cooperation to establish global norms for responsible technology use.
[179]	Quantum computing and risk prediction accuracy: an analysis of IT companies' risk appetite	This research examines how Quantum Amplitude Estimation (QAE) enhances IT risk appetite prediction, using surveys, interviews, and statistical modeling. It demonstrates QAE's impact on forecast accuracy, data security, and market volatility, offering a foundation for quantum-enhanced risk models in IT.
[180]	A New Alliance of Machine Learning and Quantum Computing: Concepts, Attacks, and Challenges in IoT Networks	This study explores the integration of quantum computing and machine learning to enhance IoT security and efficiency. It highlights how quantum machine learning can address the scale and complexity of IoT data while discussing key challenges such as hardware limitations and algorithmic complexity.
[181]	Quantum Computing for Advanced Driver Assistance Systems and Autonomous Vehicles: A Review	This study highlights the potential of quantum machine learning (QML) to overcome key limitations in ADAS technologies, such as latency, accuracy, and security. It advocates for further research into QML to enhance real-time decision-making and efficiency in autonomous vehicles.
[182]	A survey on quantum-safe blockchain security infrastructure	This research analyzes the vulnerabilities of blockchain security infrastructure to quantum computing and evaluates NIST's quantum-resistant cryptographic algorithms. It explores practical adoption of quantum-safe solutions while aiming to preserve scalability and decentralization in blockchain systems.
[183]	Barren plateaus in variational quantum computing	This study reviews the barren plateau (BP) phenomenon in variational quantum computing, where optimization landscapes become flat and untrainable at scale. It surveys current theoretical and heuristic mitigation strategies, highlighting BPs as a key challenge driving cross-disciplinary research.

5.2. Error-correcting code cryptography

Code-based cryptography derives its security from the hardness of decoding random linear codes [187]. The McEliece cryptosystem, a prominent example, relies on the syndrome decoding problem: given a received vector $y \in \mathbb{F}_q^n$ and a generator matrix G , the attacker must find an error vector $e \in \mathbb{F}_q^n$ such that $y = Gx + e$ for some codeword x .

This problem remains intractable for quantum computers, making McEliece a strong candidate for PQC. Despite its large key sizes, McEliece has been extensively studied due to its long-standing resistance to both classical and quantum attacks [188].

5.3. Multivariate Quadratic equations (MQ)

Multivariate quadratic (MQ) cryptography is a promising PQC approach [189]. The core problem involves solving systems of quadratic equations over finite fields:

$$\begin{aligned}
 f_1(x_1, x_2, \dots, x_n) &= 0, \\
 f_2(x_1, x_2, \dots, x_n) &= 0, \\
 &\vdots \\
 f_m(x_1, x_2, \dots, x_n) &= 0,
 \end{aligned} \tag{8}$$

where each f_i is a quadratic polynomial in n variables. No efficient classical or quantum algorithms are known for solving generic MQ systems, making them suitable for cryptographic applications. MQ-based schemes are primarily used in digital signatures and public-key encryption, leveraging their resistance to quantum attacks.

5.4. Hash function-driven digital signatures

Hash-based cryptographic schemes, including Merkle trees and hash-based digital signatures, represent another quantum-resistant approach to post-quantum cryptography (PQC) [190]. Their security relies on the collision resistance of cryptographic hash functions like SHA-256. In these schemes, a Merkle tree is constructed by hashing messages into leaf nodes and recursively hashing child nodes to form interior nodes, with the root serving as the public key. Signatures consist of the authentication path from a leaf to the root.

The security of hash-based signatures stems entirely from the collision resistance of the underlying hash function, making them resistant to both Shor's and Grover's algorithms. For example, the Merkle signature scheme works by hashing a message M to $H(M)$ and verifying the signature through the hashes along the Merkle tree path. Since hash-based signatures depend on collision-resistant hash functions, quantum algorithms face significant difficulty in finding two messages with identical hash values.

5.5. Obstacles to the integration of post-quantum cryptography

The Post-quantum cryptographic (PQC) systems can achieve widespread adoption, academia and industry must address several substantial challenges. One of the primary concerns with PQC algorithms is performance. Lattice-based schemes, such as Kyber and NTRU, are significantly less efficient than classical cryptographic methods in terms of key sizes and ciphertext lengths [191]. These systems often require key sizes ranging from thousands to even millions of bytes, creating resource-intensive conditions—particularly problematic for Internet of Things (IoT) devices and other constrained environments [79,192].

The increased size of cryptographic data structures in PQC solutions leads to greater bandwidth consumption and noticeably slower encryption and decryption times. As noted in [193], this results in lower efficiency compared to traditional elliptic curve or RSA-based systems. Consequently, organizations adopting PQC must contend with performance trade-offs, including slower transaction speeds, higher storage requirements, and increased power consumption.

Beyond performance limitations, the maturity of PQC algorithms remains a critical factor. Many of the proposed solutions are still in early developmental phases. Although the National Institute of Standards and Technology (NIST) is actively leading standardization efforts, these algorithms have not yet been fully validated for long-term security and resilience against evolving quantum attacks. Ongoing research focuses on side-channel vulnerabilities, implementation flaws, and emerging cryptanalytic techniques.

Deployment challenges are further compounded by the need to integrate PQC into various existing cryptographic ecosystems, such as Transport Layer Security (TLS), Virtual Private Networks (VPNs), and digital signature frameworks [194]. Transitioning to PQC requires significant modifications to legacy systems, including extensive engineering and application-level rescripting to ensure compatibility while maintaining current security assurances.

5.6. Quantum key distribution: A breakthrough in cryptographic security

According to quantum key distribution (QKD) principles, encryption keys are generated between two parties using quantum mechanics to ensure secure communication while allowing the detection of eavesdropping [12]. QKD implements the uncertainty principle described by Heisenberg, which states that measurements disturb quantum systems. The security of QKD communication arises from the fact that any eavesdropping attempt introduces measurable errors due to the disturbance of quantum states. Bennett and Brassard established BB84 as the most recognized QKD protocol, which uses polarized photons to transmit secret keys.

In the BB84 protocol, Alice prepares randomly polarized photons, and Bob measures them using one of two selected measurement bases [195]. Key material can be securely shared between Alice and Bob only if no eavesdropper—commonly referred to as Eve—intercepts the communication. If Eve attempts to measure the photons, she introduces detectable errors, enabling Alice and Bob to discard compromised bits and continue the key generation process.

Despite its strong theoretical foundation, QKD faces several practical challenges that hinder its large-scale implementation. The most significant limitation is the restricted transmission distance. Photon signals used in QKD are highly susceptible to absorption and scattering when transmitted through optical fibers or free space, leading to instability and signal loss. These effects typically limit the maximum secure transmission distance to less than 300 km.

Future quantum communication relies on the development of quantum repeaters, which can boost quantum signals without measuring them. However, this technology remains immature due to limited progress in quantum memory, entanglement distribution, and reliable quantum protocols. A major challenge to the widespread adoption of QKD is the need for a quantum internet, which requires the construction of quantum routers, repeaters, and secure quantum channels to enable long-distance transmission of quantum states.

Overcoming these hurdles involves not only advancing the underlying technologies but also making significant technical and financial investments. Moreover, current QKD systems are vulnerable to practical issues such as side-channel attacks, where information may be leaked through unintended mechanisms, and integration inefficiencies when attempting to combine quantum communications with existing classical infrastructure. Nonetheless, quantum communication research continues to advance, with promising developments in quantum repeaters, entanglement swapping, and satellite QKD. The Chinese Micius satellite, for instance, has demonstrated long-distance quantum key distribution in space, highlighting the ongoing progress toward making secure quantum communication a global reality [196].

5.7. Hybrid systems and quantum-resistant protocols

Hybrid cryptographic systems serve as a vital approach during the pre-implementation period before all organizations adopt fully quantum-resistant cryptographic standards because they protect information [197]. These systems combine classical cryptography with quantum-resistant methods to establish multiple protection layers that integrate fundamental quantum and classical features for improved security. A hybrid system implements classical cryptographic elements, starting with RSA or ECC, alongside quantum-resistant algorithms including multivariate and lattice-based schemes that form a combination of protection against existing and future quantum security threats [188].

The hybrid system can implement ECC-based digital signature and encryption methods targeted at the elliptic curve discrete logarithm problem, and combine them with lattice-based Kyber or NTRU schemes based on the Learning With Errors problem. Quantum computers would compromise only the classical components, as they lack the ability to break quantum-resistant algorithms. Mathematically, the hybrid approach can be represented as a combination of two cryptographic operations: a classical operation $E_{\text{classical}}(m)$ and a quantum-resistant operation $E_{\text{quantum}}(m)$. The resulting ciphertext C is formed as:

$$C = (E_{\text{classical}}(m), E_{\text{quantum}}(m)) \quad (9)$$

The text message m forms part of the security framework in which both classical and quantum-resistant methods work in tandem to ensure system security. The security of messages relies upon a dual approach to encryption, where quantum-resistant protocols protect the information even if attackers manage to compromise a first layer of security (e.g., using Shor's algorithm to factorize RSA keys).

The implementation of quantum-resistant communication protocols starts with NTRUEncrypt and Rainbow, as well as other cryptographic methods built on lattice-based and multivariate quadratic equations [198–200]. These solutions represent an optimistic path toward quantum-secure communication systems. A quantum-resistant system like Kyber and other lattice-based encryption schemes relies on the Module Learning With Errors (MLWE) problem, a computational challenge involving the identification of short vectors in high-dimensional lattices—a task that Shor's algorithm cannot effectively address [201]. Key exchange operations in cyberspace utilize polynomial rings to enable secure secret exchanges between two parties, while the ring-LWE problem provides the security guarantee for this process. Mathematically, the encryption in Kyber can be expressed as:

$$C = A \cdot r + e \quad (10)$$

where A is the public key, r is a random vector, and e is a noise term added to obscure the original message. The computational intractability of the LWE problem ensures that even with quantum computers, an attacker cannot easily decipher the key.

Rainbow protocols use multivariate quadratic equations to generate digital signatures by solving systems of polynomial equations over finite fields, which are considered quantum-resistant [202]. The multivariate approach for digital signatures utilizes, which demonstrates a system of quadratic equations as a basis for its operation, using n quadratic polynomials f_i . The computational difficulty of Rainbow's systems ensures security, as no known efficient solution exists for these problems even with quantum computers. The quantum search algorithm developed by Grover, along with its analogues, shows no significant acceleration in solving multivariate systems, thus maintaining the protection of digital signatures [203].

Hybrid systems not only protect existing infrastructures against quantum threats but also lay the foundation for fully post-quantum secure architectures by using and combining quantum-resistant techniques like lattice-based encryption (Kyber) [204], multivariate-based signatures (Rainbow), and other quantum-safe primitives. These hybrid systems are necessary for secure communication during the transition period because they maintain encryption security even when part of the system is compromised by quantum computing. However, this challenge is not yet fully addressed, especially in terms of efficiency and seamless integration with existing cryptographic infrastructures—particularly given the larger key sizes and computational overhead associated with quantum-resistant algorithms [12].

In Table 9, we explore innovative mitigation strategies to counteract the threats of quantum computing in cybersecurity, offering a wide range of solutions tailored to specific challenges. For instance, Quantum Random Number Generators (QRNGs) ensure high entropy for secure cryptographic keys, while Quantum Safe DNSSEC aims to reinforce the foundational internet infrastructure against quantum attacks. Cutting-edge concepts such as quantum-enhanced honeypots and quantum-powered cyber threat intelligence showcase proactive approaches to detecting and mitigating quantum-enabled threats. Meanwhile, advancements in communication—such as Quantum Tunneling Secure Channels and Quantum Internet Protocol Development—highlight efforts to future-proof secure communications using quantum mechanics. Additionally, solutions like Post-Quantum Data Sanitization and Dynamic Key Evolution Mechanisms address long-term data security and adaptive cryptographic defenses. However, implementing these strategies requires overcoming challenges like high costs, computational overhead, and the need for global standardization and cooperation, underscoring the complexity of securing a quantum future.

6. Emerging solutions and current research progress

6.1. Quantum-resistant cloud security

The future protection of cloud data faces risks from quantum computers that could effectively break the prevalent encryption security protocols currently in use for protection purposes [205]. Cloud service providers protect transit and rest data security through postquantum cryptographic (PQC) solutions based on traditional encryption techniques. The approach aims to make

Table 9

Concise mitigations for quantum threats in cybersecurity.

Mitigation	Description	Pros and Cons
Quantum RNGs	Use quantum phenomena to generate truly random cryptographic keys.	High entropy; needs specialized hardware.
Quantum-safe DNSSEC	Apply PQ crypto to secure DNS infrastructure.	Strengthens trust; hard to deploy globally.
PQ zero-knowledge proofs	Enhance ZKPs with quantum-safe primitives.	Secures authentication; resource-intensive.
Quantum honeypots	Trap attackers using quantum protocols.	Detects threats early; costly to implement.
Quantum tunneling channels	Use tunneling for ultra-secure comms.	Highly secure; experimental tech.
PQ voting systems	Merge PQ crypto with blockchain for secure voting.	Tamper-proof; needs public/regulatory trust.
Quantum threat intel	Accelerate threat detection with quantum algorithms.	Fast insights; limited scalability.
PQ certificate authorities	Shift CAs to post-quantum standards.	Secures PKI; needs global consensus.
Quantum data sharding	Split sensitive data across locations with quantum methods.	More secure storage; harder access control.
AI-PQ defense	Optimize PQ crypto with AI support.	Efficient; AI introduces new risks.
Quantum internet protocols	Build secure comm protocols for quantum internet.	Future-ready; still experimental.
Dynamic key evolution	Use quantum randomness for evolving keys.	Real-time security; heavy computation.
PQ autonomous systems	Use PQ encryption in drones/vehicles.	Secures comms; hardware limitations.
PQ data sanitization	Encrypt and erase data securely for the quantum era.	Strong privacy; hard to scale.
Global PQ frameworks	Coordinate international PQ crypto standards.	Promotes collaboration; slow to align.

quantum computers less effective against the commonly used RSA or ECC Mailbox Procedures by implementing Shor's algorithm-linked frameworks [206], which break large integers and discrete logarithms using polynomial-time factoring. Hybrid cryptographic strategies represent a solution to deal with the transition to quantum-resistant methods by combining quantum-resistant encryption with existing traditional encryption systems until quantum computing reaches practical security attack capabilities [207].

When hybrid systems are implemented, the same AES encryption technology pairs with asymmetric RSA/ECC [208], alongside quantum-resistant algorithms such as lattice-based cryptography (Kyber for key exchange and NTRU for encryption) and hash-based signatures (XMSS) to generate digital signatures [209].

Furthermore, cloud providers are incorporating quantum key distribution (QKD) techniques, particularly to secure communications between users and cloud servers [210]. QKD, as demonstrated by protocols such as BB84, uses quantum mechanics principles to securely exchange keys, making it fundamentally resistant to eavesdropping [195]. The quantum key-exchange process can be mathematically described as:

$$K_{\text{shared}} = \text{QKD}(\mathbf{A}, \mathbf{B}, E) \quad (11)$$

The shared secret key K_{shared} , quantum states $\mathbf{A}$ and $\mathbf{B}$ used by Alice and Bob respectively, and the eavesdropping detection capability E ensure that any interception is revealed through disturbances in quantum system measurement. The practical application of QKD faces significant barriers because it requires the development of a quantum Internet, along with overcoming the natural limitations of quantum states such as transmission distance, loss, and noise. To overcome these, cloud providers propose merging quantum communication protocols into existing infrastructure using quantum repeaters and satellite-based communication systems to expand the reach and scale of the QKD network, according to [211].

Moreover, quantum-resistant cryptography is being incorporated into cloud storage systems to protect data at rest [212]. Post-quantum algorithms, such as Kyber (for key exchange) and NTRU (for encryption), are implemented to ensure that data remain secure even if future quantum computers can efficiently break traditional systems [213]. The principle behind lattice-based encryption schemes such as Kyber depends on the Learning With Errors (LWE) problem. Specifically, where $\mathbf{A}$ is public, r is random, and $\mathbf{e}$ is noise, to ensure that the secret decryption key cannot be feasibly derived, even with quantum computing. In the short and medium term, hybrid cryptographic systems are needed to guarantee the protection of cloud data until quantum-safe algorithms can be fully deployed throughout cloud infrastructures.

Cloud providers are developing post-quantum secure hybrid cloud architectures that unite quantum-resistant cryptographic protocols with classical security layers to provide complete end-to-end protection, according to [214]. They aim to incorporate lattice encryption into storage systems and use multivariate signatures for authentication, combined with hash functions for security verification within existing frameworks. Hybrid systems combine classical and quantum-resistant algorithms into one integrated security model for dual cloud environments to defend against both classical and quantum attackers, according to [215].

Finally, we conclude that quantum-resistant cloud security strategies leverage hybrid cryptography and quantum-resistant algorithms to enable the continued protection of cloud services as quantum computing advances [216]. Cloud providers face challenges in making their systems performant and scalable, and in using PQC algorithms that are sufficiently mature. However, they are on the path to providing strong security against future quantum threats, ensuring that data remain confidential, verifiable, and private in the quantum world.

6.2. Quantum-safe blockchain

The exploration of quantum-safe blockchain technology identifies vulnerabilities in blockchain systems related to quantum computing operations, as demonstrated in research (e.g., [203]). Blockchain systems rely on cryptographic primitives such as public-key cryptography and digital signatures for authentication, yet they remain vulnerable to future quantum computing capabilities.

Current cryptographic methods used in blockchains depend on mathematical problems like Integer Factorization (which secures RSA) and the Discrete Logarithm Problem (which secures ECC). These methods can be efficiently solved by quantum algorithms, particularly Shor's algorithm, which enables attackers to forge signatures and compromise public-key infrastructures [217].

The security of ECC-based blockchain systems hinges on the hardness of the elliptic curve discrete logarithm problem, where solving $P = kQ$ (with P and Q being points on the elliptic curve and k an unknown scalar) must remain computationally infeasible. However, Shor's algorithm can solve this problem in polynomial time, thus threatening the security of ECC-based signatures in blockchain systems.

To mitigate these risks, quantum-resistant cryptography should be integrated into blockchain protocols to ensure security even against quantum-capable attackers [218]. Lattice-based cryptography, particularly schemes based on the Learning with Errors (LWE) problem, offers a promising solution. For instance, Kyber (for key exchange) and FrodoKEM (for encryption) leverage LWE's quantum resistance, as quantum algorithms fail to achieve significant speedups in solving lattice problems [77,219]. The Kyber key exchange process can be mathematically represented as:

$$\mathbf{A}s + \mathbf{e} = \mathbf{b} \quad (12)$$

Here, $\mathbf{A}$ is a public matrix, s is the secret message, $\mathbf{e}$ is a noise vector, and $\mathbf{b}$ is the encrypted output. The security of lattice-based cryptography relies on the hardness of finding short vectors in high-dimensional lattices, a problem resistant to both classical and quantum attacks. Adopting such methods in blockchain protocols can safeguard public-key exchanges and encryption systems against quantum threats.

Hash-based cryptography (e.g., XMSS or SPHINCS+) provides another quantum-resistant approach through one-time signatures that exploit the collision resistance of cryptographic hash functions [220]. These signatures remain secure against quantum attacks because quantum algorithms do not significantly accelerate finding hash collisions. XMSS constructs a hash tree, with the root node serving as the public key and leaf nodes derived from hashing random values [221]. The security of XMSS depends on the collision resistance of its underlying hash function, even against quantum computers. The signing process in XMSS can be expressed as:

$$S_i = H_i(\mathbf{m}, \mathbf{r}_i) \quad (13)$$

Here, S_i is the signature for message $\mathbf{m}$, generated using hash function H_i and a random value $\mathbf{r}_i$. Hash-based signatures are particularly appealing for blockchains due to their proven quantum resistance.

However, adopting quantum-resistant cryptography in blockchains faces several challenges [222]. For instance, lattice-based schemes often require larger key sizes than RSA or ECC, increasing transaction sizes and verification times, which may impact blockchain scalability and performance. This dissertation explores hybrid systems that combine classical cryptography (e.g., RSA/ECC) with quantum-resistant techniques (e.g., lattice-based or hash-based cryptography) to provide transitional security until quantum-safe algorithms mature [190].

Quantum key distribution (QKD) can further enhance blockchain security by enabling eavesdropping-resistant key exchange between nodes [223]. QKD's ability to detect eavesdropping ensures secure key reestablishment, preventing unauthorized access [224]. Mathematically, the QKD key exchange can be represented as:

$$K_{\text{shared}} = f(\mathbf{A}, \mathbf{B}) \quad (14)$$

Here, K_{shared} is the shared secret key derived from the quantum states $\mathbf{A}$ (Alice) and $\mathbf{B}$ (Bob), with any eavesdropping attempt E causing detectable disturbances. Integrating QKD with blockchain networks provides a quantum-secure communication layer.

In conclusion, blockchain protocols must adopt quantum-resistant solutions like lattice-based cryptography and hash-based signatures to counter future quantum threats [225]. Hybrid cryptographic approaches can provide interim security while transitioning to fully quantum-resistant infrastructures [226]. However, addressing performance, scalability, and interoperability challenges is critical for real-world deployment, ensuring decentralized networks remain secure in the quantum era [209].

6.3. Secure and lightweight authentication schemes for cloud and healthcare systems

Progress on authentication scheme for cloud computing based on Chebyshev chaotic maps [227]. The model enhances security by supporting features like mutual authentication, two-factor authentication, password flexibility, and forward security—even if the server's private key is compromised. It also ensures user anonymity and untraceability. The scheme is proven secure under the Diffie–Hellman assumption and performs better than existing methods against various attacks in cloud environments.

In study [228], authors introduce a lightweight and privacy-preserving user authentication scheme for smart healthcare systems using Big Data. It aims to prevent unauthorized access to sensitive health data and control over IoT-enabled medical devices. The proposed method ensures only legitimate users can securely access and modify patient data, protecting against malicious tampering. Security analysis confirms that the scheme is both efficient and more secure than existing solutions.

Finally, an efficient Revocable Identity-Based Signature (RIBS) scheme to enhance data security in cloud environments [229]. It tackles the challenge of revoking malicious users by offloading revocation tasks to an External Revocation Server (ERS), reducing the burden on the Key Generation Center. Only non-revoked users can access system resources. The scheme uses a time-based key for signing messages and combines symmetric encryption with ECC-based encryption to ensure data confidentiality. Results show improved performance and lower computation costs compared to existing methods.

6.4. Quantum cryptography in IoT and edge computing

Quantum cryptography in the context of the Internet of Things (IoT), where it is applied to edge computing, is an important frontier for protecting against classical and quantum attacks on highly distributed low-power devices [230]. The IoT ecosystem, consisting of billions of interconnected sensors, devices, and edge nodes, demands solutions that provide robust security while respecting the stringent constraints of power, computational resources, and latency [231,232].

Quantum cryptographic techniques, including quantum key distribution (QKD) and quantum-resistant encryption schemes, provide promising means to achieve secure communication and data protection in such environments [128]. As such, protocols such as the BB84 QKD scheme allow cryptographic keys to be securely exchanged between devices without the possibility of interception, since any eavesdropping on a quantum communications channel automatically disrupts the quantum system, and the communication parties can detect it [233].

In mathematical terms, the quantum cryptographic key is transmitted by sending quantum states $|\psi\rangle$. The key exchange security in QKD is based on the quantum no-cloning theorem, which states that an unknown quantum state cannot be perfectly cloned; hence, eavesdropping is detectable. For example, in the BB84 protocol, Alice sends a series of quantum bits (qubits) in one of four possible polarization states, say $|\pm 0\rangle, |\pm 1\rangle$, along the X and Z -bases, measuring the qubits and distilling the shared key from the measurement outcomes.

The main problem in IoT and edge computing environments is related to the limited resources of IoT devices, which have low memory, power, and computational capabilities [234]. The resources on these devices are insufficient to directly perform standard QKD protocols, yet standard QKD systems require a precise optical setup and significant processing capabilities to deal with the qubit state. Integrated hybrid quantum-classical systems are being investigated as a solution, where the quantum layer handles key exchange, while the classical layer is responsible for data encryption using post-quantum algorithms, such as lattice-based encryption schemes (e.g., Kyber for key exchange or NTRU for encryption) [235]. As an example, Kyber relies on the quantum-resistant Learning With Errors (LWE) problem and achieves competitive key sizes and low computational overhead.

This provides an effective means for IoT devices and edge nodes to securely exchange keys, avoiding traditional quantum-vulnerable cryptography. In addition, certain hash-based cryptographic schemes (e.g., XMSS) offer quantum resistance, utilizing hash functions that are hard to invert and one-time signatures that are difficult to forge even with quantum computing. The XMSS scheme works by applying a hash function H to generate a one-time signature, such that in S_i is the signature for $\mathbf{m}$, generated using a hash function H_i on the message $\mathbf{m}$ and the random value $\mathbf{r}_i$. However, in the quantum era, this method remains secure since quantum algorithms do not provide a significant advantage in finding preimages or collisions in hash functions. Such quantum-resistant algorithms are therefore essential for IoT and edge computing adoption, ensuring data integrity, confidentiality, and authentication in an environment where standard cryptographic schemes may be vulnerable to quantum attacks.

Quantum key distribution (QKD) and quantum-resistant cryptography significantly impact future IoT and edge devices, making them critical considerations for researchers [214]. Although low-power sensor devices are deployed across extensive locations to extend QKD reach, quantum-safe networking protocols are still required [236]. The only viable method for achieving this involves quantum repeaters and quantum communication networks, which enable quantum key distribution over longer distances than direct QKD [237]. This paves the way for a future quantum internet, where these novel technologies can enable quantum-secure communication for IoT devices revolutionizing edge computing security. In such a system the security of the quantum key exchange process ensures that even in the event of a physical compromise or operation in an untrusted environment the integrity of the exchanged keys and sensitive data remains protected [164].

Quantum cryptography serves as a potential solution to protect devices with restricted power and limited resources by safeguarding them against the impending threat of quantum computing [238]. The same IoT networks can deliver quantum-resistant encryption and hash-based signature capabilities after performing secure key exchange with QKD solutions. The implementation of these technologies faces three practical difficulties stemming from resource constraints along with the distance limitations of communication and the need to integrate quantum technologies with existing network infrastructure. The next generation of distributed and interconnected devices will need quantum-safe IoT and edge computing systems because the hurdles to implementing advanced cryptanalytic computing have been resolved [230].

6.5. Quantum internet

The emerging field of quantum Internet research seeks to build secure communication networks which use quantum mechanics principles, especially quantum entanglement with teleportation, to provide fundamental network security [239]. Both classical communication systems and the quantum Internet transmit information differently because classical systems use classical bits, while quantum networks use quantum bits (qubits), which present a superposition state to enable powerful data transmission methods [240]. The essential quantum communication trait relies upon entanglement, which describes the mutual yet distant connection between two qubits through their synchronized states. The mathematical description of entanglement uses the Bell state to describe it:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (15)$$

The two qubits in an entangled state exist only in the basis states $|00\rangle$ and $|11\rangle$. Scientifically proven pairs of qubits become unbreakably linked after their entanglement process is completed because their states immediately change upon mutual measurements, and this effect extends between distant qubits, making quantum channels highly resistant to eavesdropping since

any patterns of disruption reveal monitoring activity. The QKD protocol BB84 depends on quantum channels to distribute secret keys based on this principle. The mathematical security aspect of QKD results from combining the no-cloning theorem with the uncertainty principle, thus preventing copies of qubits and keeping them invisible to measurement before state change [241].

Quantum teleportation enables remote data exchange between two qubits through quantum entanglement because it transfers quantum information without sending qubits individually [242]. Two parties utilizing the teleported information need to have an entangled pair, which serves as the protocol foundation. When Alice desires to teleport a quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob, she applies a Bell state measurement on her entangled pair and the state $|\psi\rangle$, which results in one of the four Bell states forming. Bob receives Alice's classical measurement results, through which he applies an appropriate unitary operation to transform his entangled pair into the precise state $|\psi\rangle$. When implemented mathematically, the process consists of these three consecutive steps:

1. Alice has a quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to be teleported and an entangled pair in the Bell state $|\Phi^+\rangle$:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (16)$$

2. Alice performs a measurement of the Bell state in the system $|\psi\rangle \otimes |\Phi^+\rangle$, leading to the collapse of the combined state into one of the four Bell states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|\psi\rangle_{AB} + |\psi\rangle_{A'B'}) \quad (17)$$

3. Bob, however, applies the unitary transformation U (such as the Pauli-X or Pauli-Z gates) to his half of the entangled state using the measurement information sent to him through a classical channel by Alice, resulting in the recovery of the original state $|\psi\rangle$.

The shared pair of entangled states between Alice and Bob allows instantaneous quantum information transfer at any distance without the risk of interception, thus establishing secure communication. Yet, there already exist substantial challenges in the practical realization of quantum teleportation and entanglement at large scales, particularly in quantum repeater networks for quantum communication beyond the coherence length of entangled photons. A quantum repeater serves as an entanglement-swapping intermediary that extends the distance over which quantum entanglement may be maintained. Entanglement swapping mathematically means that we create entanglement between two qubits that have never directly interacted by measuring a Bell state of two intermediate entangled pairs, such that entanglement is created between far-away qubits [243].

One proposed method for creating a quantum Internet is the quantum repeater network, where quantum repeaters are deployed to enable long-distance entanglement and teleportation across a large-scale quantum communication network [244]. These repeaters rely on entanglement purification protocols, which improve the fidelity of entanglement by discarding low-quality entangled states and keeping only high-fidelity ones. Entanglement purification, mathematically, is represented as a probabilistic process where multiple low-fidelity entangled states are combined to produce fewer, but higher-fidelity, entangled states.

$$|\Phi_{\text{purified}}\rangle = \text{Entanglement Purification}(|\Phi_{\text{noisy}}\rangle) \quad (18)$$

The execution of research aims to address crucial operational problems such as quantum error correction because this mechanism safeguards quantum state transmission, as states lost during transmission lead to decoherence, which weakens quantum Internet security and dependability [106]. The Shor code and the Steane code function as quantum error correction schemes because they encode quantum information into several qubits through entangled states.

Protected communication on the quantum Internet is based on its quantum signature features of entanglement and teleportation to detect intercept attempts, making them easily detectable according to [245]. A technological system which derives its security from quantum mechanics, along with entanglement theory, the no-cloning theorem, and quantum error correction protocols, protects data during transmission. A quantum Internet seems possible for futuristic global communication systems because it provides protection that classical security methods cannot match, despite technological challenges with reach extension, quantum repeater development, and the realization of a larger network [128].

7. Comparative analysis between quantum computing and cybersecurity

The Table 10, provides a comprehensive comparison of various Post-Quantum Cryptography (PQC) algorithms based on their security and performance metrics. It evaluates six prominent algorithms: Kyber, NTRU, McEliece, XMSS, Rainbow, and SIDH, each categorized by their underlying security basis, such as Module-LWE for Kyber, Ring-LWE for NTRU, Syndrome Decoding for McEliece, Hash-based (SHA-256/3) for XMSS, Multivariate Quadratics for Rainbow, and Supersingular Isogenies for SIDH. The table highlights key attributes like key size (ranging from 1568 bits for Kyber to over 1 MB for McEliece), latency (from 0.1 ms for XMSS to 100 ms for McEliece), and NIST security levels (1–5, where higher levels indicate stronger security). Additionally, it assesses hardware compatibility, noting that Kyber and NTRU are suitable for CPUs and GPUs, while McEliece excels in GPU-parallel environments, and XMSS is optimized for IoT/edge devices. The standardization status is also included, with Kyber and XMSS being NIST-standardized, while Rainbow was rejected, and SIDH was broken due to the GPST attack. Known vulnerabilities, such as side-channel attacks for Kyber (mitigated), key mismatch for NTRU, and the UOV attack for Rainbow, are also documented, providing a holistic view of each algorithm's robustness and limitations in real-world applications.

The Table 11, presents a comparative analysis of four prominent Quantum Key Distribution (QKD) protocols—BB84, E91, TF-QKD, and CV-QKD—based on key performance and operational metrics. Each protocol is distinguished by its underlying security mechanism, such as polarization encoding for BB84, entanglement-based EPR pairs for E91, twin-field methodology for TF-QKD,

Table 10

Comparison of Post-Quantum Cryptography (PQC) algorithms based on security and performance metrics.

Algorithm	Security basis	Key size (bits)	Latency (ms)	NIST level	Hardware friendliness	Standardization status	Known attacks
Kyber (Lattice)	Module-LWE	1568–3168	0.5–2.0	1–5	CPU/GPU/FPGA	NIST Std	Side-channel (mitigated)
NTRU	Ring-LWE	2048–4096	1.0–5.0	1–3	CPU/GPU	NIST Finalist	Key mismatch
McEliece	Syndrome decoding	1 MB+	10–100	1–5	GPU (parallel)	NIST Finalist	None (structural)
XMSS (Hash)	SHA-256/3	2048–10,000	0.1–1.0	1–3	IoT/Edge	NIST Std.	Pre-image (theoretical)
Rainbow (MQ)	Multivariate quadratics	100–200 kB	5–20	1–3	CPU	Rejected	UOV attack
SIDH	Supersingular isogenies	564–1008	2–10	1–3	CPU	Broken	GPST attack

Table 11

Quantum Key Distribution (QKD) protocol comparison.

Protocol	Security basis	Max distance (km)	Key rate (Mbps)	Decoherence tolerance	Cost	Commercial readiness
BB84	Polarization encoding	300 (fiber)	0.1–10	Low	\$\$\$\$	Mature (ID Quantique)
E91	Entanglement (EPR)	100 (fiber)	0.01–1	Medium	\$\$\$\$\$	Lab trials (China)
TF-QKD	Twin-Field	500+ (fiber)	0.5–5	High	\$\$\$\$\$	Deployed (Huawei)
CV-QKD	Continuous variables	50 (free space)	1–100	Low	\$	SK Telecom

Table 12

Comparison of cryptographic primitives under quantum threat.

Metric	RSA-2048	ECC-256	AES-256	Kyber-1024	XMSS-SHA256
Quantum attack time	Minutes (Shor)	Seconds (Shor)	2^{128} (Grover)	None	None
Key size (bits)	2048	256	256	2400	2048
Energy/Op (μ J)	50–100	10–20	0.1–1	5–10	0.5–2
IoT suitability	No	Yes	Yes	Partial	Yes

and continuous variables for CV-QKD. The table highlights practical deployment considerations, including maximum achievable distances (ranging from 50 km for CV-QKD in free space to over 500 km for TF-QKD in fiber), key generation rates (from 0.01 Mbps for E91 up to 100 Mbps for CV-QKD), and tolerance to decoherence (with TF-QKD exhibiting high resilience, while BB84 and CV-QKD are more susceptible). Cost implications are also outlined, with BB84 and CV-QKD being relatively more affordable (denoted by \$–\$\$\$\$), whereas E91 and TF-QKD are high-cost (\$\$\$\$–\$\$\$\$\$) due to their advanced requirements. Commercial readiness is assessed, noting BB84’s maturity with deployments by firms like ID Quantique, TF-QKD’s real-world implementation by Huawei, CV-QKD’s adoption by SK Telecom, and E91’s current limitation to laboratory trials, primarily in China. This comparison underscores the trade-offs between distance, speed, robustness, and economic feasibility in QKD technologies.

The [Table 12](#), compares the vulnerability and performance of five cryptographic primitives—RSA-2048, ECC-256, AES-256, Kyber-1024, and XMSS-SHA256—in the context of quantum computing threats. It evaluates them across four key metrics: resistance to quantum attacks, key size, energy efficiency, and suitability for IoT applications.

Under quantum attack time, classical algorithms like RSA-2048 and ECC-256 are shown to be highly vulnerable, with Shor’s algorithm breaking them in minutes and seconds, respectively. In contrast, AES-256 retains some security due to Grover’s algorithm only providing a quadratic speedup (reducing brute-force complexity to 2^{128} operations). Post-quantum alternatives like Kyber-1024 (a lattice-based scheme) and XMSS-SHA256 (a hash-based signature scheme) are marked as having no known efficient quantum attacks.

Key size comparisons reveal that Kyber-1024 has the largest keys (2400 bits), while ECC-256 and AES-256 are the most compact. Energy consumption per operation varies significantly, with AES-256 being the most efficient (0.1–1 J), followed by XMSS-SHA256 (0.5–2 J), while RSA-2048 is the least efficient (50–100 J).

Finally, IoT suitability is assessed, with ECC-256 and AES-256 being well-suited due to their efficiency, XMSS-SHA256 also qualifying despite higher computational demands, and Kyber-1024 only partially suitable due to its larger key sizes and higher energy costs. RSA-2048 is deemed impractical for IoT. This table highlights the urgent need for quantum-resistant cryptography while illustrating trade-offs in key size, energy use, and deployment feasibility.

The [Table 13](#), compares four blockchain platforms that have integrated post-quantum cryptography (PQC) to enhance their security against quantum computing threats. Each blockchain is evaluated based on its chosen PQC method, transaction processing speed (TPS), consensus mechanism, quantum resilience level, and current adoption status.

Quantum Resistant Ledger (QRL) employs the hash-based XMSS signature scheme, providing high quantum resistance but with a relatively modest TPS of 100 due to its proof-of-stake (PoS) consensus. It has been operational on mainnet, making it one of the earliest adopters of PQC in blockchain.

Table 13

PQC-enabled blockchain solutions and characteristics.

Blockchain	PQC method	TPS	Consensus	Quantum resilience	Adoption status
QRL	XMSS	100	PoS	High (Hash-based)	Mainnet
Hyperledger	Kyber	1000+	BFT	High (Lattice)	Pilot
Algorand	Falcon	5000	Pure PoS	High (Lattice)	Proposed
IOTA	Winternitz	500	DAG	Medium (OTS)	Testnet

Table 14

Hardware platform constraints for PQC and QKD deployment.

Platform	PQC support	QKD support	Power (W)	Latency (ns)	Cost (USD)
CPU (x86)	Kyber, Falcon	No	10–100	50–200	\$100–\$1000
FPGA	All PQC	BB84	5–20	10–50	\$1000–\$5000
IoT (ARM)	XMSS, SPHINCS+	No	0.1–1	100–500	\$10–\$100
QKD Chip	No	BB84/TF-QKD	1–5	1–10	\$10,000+

Hyperledger implements Kyber (a lattice-based key encapsulation mechanism) and achieves over 1000 TPS using Byzantine Fault Tolerance (BFT) consensus. Its lattice-based approach ensures strong quantum resistance while maintaining high throughput.

Algorand has proposed integrating Falcon, another lattice-based signature scheme, by 2024. With a pure PoS consensus, it boasts an impressive 5000 TPS, positioning it as a high-performance quantum-resistant blockchain solution.

IOTA, which uses a directed acyclic graph (DAG) structure instead of traditional blockchain, adopts the Winternitz one-time signature (OTS) scheme. While it offers medium quantum resilience (due to potential limitations of OTS), it maintains a balance with 500 TPS and is currently in testnet phase.

This table highlights the trade-offs between quantum security, scalability, and deployment maturity, illustrating how different blockchain platforms are preparing for the post-quantum era.

The Table 14, provides a detailed comparison of hardware platforms in terms of their suitability for deploying Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD) solutions. It evaluates four distinct hardware platforms—general-purpose CPUs (x86), Field-Programmable Gate Arrays (FPGAs), IoT devices (ARM-based), and specialized QKD chips—across five critical metrics: PQC support, QKD support, power consumption, operational latency, and cost.

For PQC support, the table shows that x86 CPUs can efficiently run lattice-based algorithms like Kyber and Falcon, while FPGAs offer the most flexibility by supporting all PQC algorithms. IoT devices, with their constrained resources, are limited to hash-based schemes (XMSS, SPHINCS+). In contrast, QKD chips are specialized for quantum key distribution and do not support PQC.

When it comes to QKD support, only FPGAs and dedicated QKD chips are capable of running QKD protocols, with FPGAs supporting BB84 and QKD chips supporting both BB84 and the more advanced TF-QKD. General-purpose CPUs and IoT devices lack QKD compatibility entirely.

Power consumption varies significantly across platforms, ranging from ultra-low power (0.1–1 W) for IoT devices to moderate (5–20 W) for FPGAs, and higher (10–100 W) for CPUs. QKD chips, while power-efficient (1–5 W), come at a premium cost (\$10,000+).

Latency is lowest for QKD chips (1–10 ns) and FPGAs (10–50 ns), making them suitable for real-time applications. CPUs and IoT devices exhibit higher latency (50–500 ns), with IoT devices being the slowest due to their limited processing capabilities.

Finally, cost considerations reveal that IoT devices are the most economical (\$10–\$100), followed by CPUs (\$100–\$1000) and FPGAs (\$1000–\$5000). QKD chips, while offering the best performance for QKD, are prohibitively expensive (\$10,000+), limiting their deployment to high-security, high-budget scenarios.

This table underscores the importance of selecting the appropriate hardware platform based on specific use-case requirements, balancing factors such as cryptographic support, power efficiency, speed, and cost.

In Fig. 3, we presents a comparative analysis of three cybersecurity paradigms—Traditional Cybersecurity, Modern Cybersecurity, and Quantum Computing—across ten key attributes: Encryption Strength, Vulnerability to Attacks, Scalability, Cost Efficiency, Adoption Rate, Energy Efficiency, Attack Resilience, Key Size, and Latency. Each attribute is represented as a percentage on the y-axis, reflecting its relative effectiveness or performance within each paradigm, while the x-axis lists the categories being compared. Quantum Computing demonstrates superior Encryption Strength and Attack Resilience (near 100%) due to its theoretical unbreakability with post-quantum cryptography, but it lags in Scalability, Cost Efficiency, and Adoption Rate (lower percentages) due to its nascent stage, high implementation costs, and limited real-world deployment. Modern Cybersecurity strikes a balance, outperforming Traditional Cybersecurity in Vulnerability to Attacks and Energy Efficiency due to advanced protocols like elliptic-curve cryptography and optimized algorithms, while Traditional Cybersecurity remains more Cost-Efficient and Scalable due to its maturity and widespread infrastructure.

The percentage values quantify the relative advantage or disadvantage of each paradigm in a given category—higher percentages indicate stronger performance (e.g., 90% in Encryption Strength for Quantum Computing implies near-ideal security), while lower percentages (e.g., 40% in Adoption Rate for Quantum Computing) highlight current limitations. The discussion underscores that while Quantum Computing excels in future-proof security, its practical challenges (e.g., latency, key size overhead) necessitate

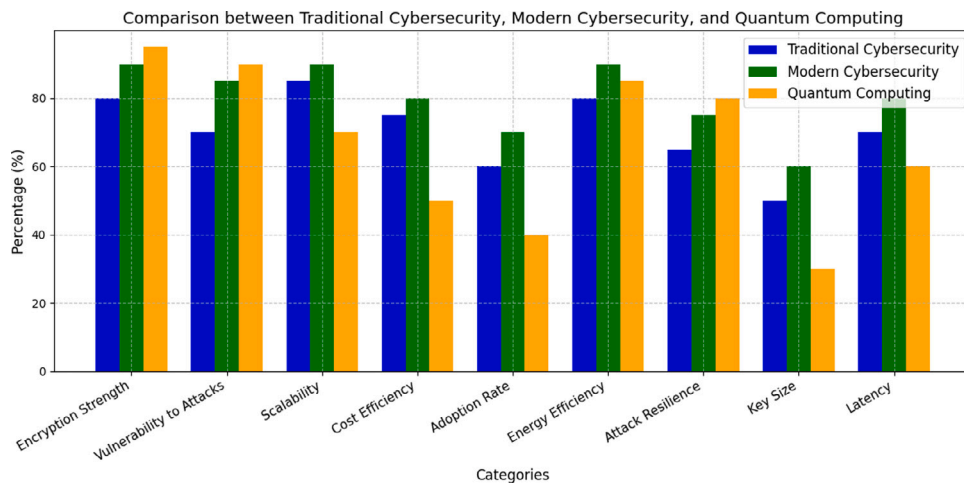


Fig. 3. Comparison between traditional cybersecurity, modern cybersecurity, and quantum computing.

Table 15

Summary of reviewed articles comparison.

References	Article type	Proposed method	Limitations
Baseri, Yaser, et al. (2024) [246]	Review	Quantum-resistant security framework with tailored defenses for nine critical infrastructure components and strategic countermeasures	Uncertain timeline for quantum threats complicates prioritization of mitigation
Jowarder, Rafiul Azim, et al. (2024) [247]	Review	Examines quantum computing threats to cryptography and suggests post-quantum solutions	Timeline for practical quantum computing and transition challenges
Mangla, Cherry, et al. (2023) [248]	Review	Quantum-based security for 5G vulnerabilities and robust 6G network protection	Theoretical focus with limited practical applicability
Faruk, Md Jobair Hossain, et al. (2022) [249]	Review	Overview of quantum computing's threat and enhancement to cybersecurity	Limited reviewed studies, lacking algorithmic detail
Moura, Luiz Filipi, et al. (2025) [250]	Review	Focus on Quantum Key Distribution (QKD) for strong quantum security	Lack of comparative and mathematical details
Ahmed, Ummar, et al. (2024) [251]	Review	Quantum computing applications enhancing cybersecurity across domains	Lack of comparison and algorithmic details
Our proposed work	Review	Evaluation of quantum threats and emerging quantum-safe cybersecurity solutions	Lack of real-world implementation

hybrid approaches during the transition period, whereas Modern Cybersecurity offers a pragmatic middle ground with improved defenses without the radical overhaul required by quantum systems.

Finally, we have summarized several reviewed articles that are most relevant to our proposed work, as presented in Table 15. Readers can refer directly to this comparison for further insight.

8. Challenges and future directions: Quantum computing on cybersecurity

Quantum computing poses serious risks to current cryptographic systems, especially public-key algorithms like RSA and ECC, which are vulnerable to Shor's algorithm. Similarly, Grover's algorithm threatens symmetric encryption by halving effective key lengths, demanding larger keys and more resources. These developments highlight the urgent need for post-quantum cryptography (PQC).

Lattice-based, code-based, hash-based, and multivariate cryptographic schemes offer promising quantum resistance, with the Learning With Errors (LWE) problem forming the backbone of many secure approaches. However, PQC adoption faces major challenges: increased key sizes, computational overhead, and limited algorithmic maturity hinder deployment in resource-constrained environments such as IoT. Further concerns include resistance to side-channel attacks and the lack of interoperability with existing infrastructure.

Hybrid cryptographic models are crucial during the transition phase, allowing classical and quantum-safe algorithms to coexist. Scalability of quantum key distribution (QKD) and the development of robust quantum networks also remain immature, facing

Table 16

Challenges and future directions in quantum cybersecurity.

Area	Challenge	Future direction
Cryptographic systems	Quantum algorithms break RSA, ECC, AES.	Adopt post-quantum and hybrid cryptographic schemes.
Performance	PQC needs more resources and larger keys.	Optimize PQC for speed, especially in IoT/mobile.
Algorithm maturity	PQC is still evolving; some are unproven.	Validate through deployment and testing.
Interoperability	Legacy systems incompatible with PQC.	Use hybrid crypto for gradual integration.
Quantum key distribution	Limited by distance, photon loss.	Develop quantum repeaters and error correction.
Quantum-safe blockchain	ECDSA vulnerable in quantum era.	Use lattice- or hash-based signatures.
Threat detection	Existing systems cannot detect quantum threats.	Design quantum-aware detection frameworks.
Cloud security	Encryption and data integrity at risk.	Embed PQC and hybrid models in cloud services.
IoT	PQC too heavy for constrained devices.	Create lightweight PQC for edge/IoT.
Quantum internet	Challenges in entanglement and scaling.	Improve repeaters, decoherence control, and routing.
Quantum networks	Decoherence and loss limit scalability.	Advance entanglement distribution tech.
Attack prevention	Quantum attacks bypass current defenses.	Build quantum-resilient security tools.
Ethics	Raises governance and surveillance issues.	Develop global policy and ethical frameworks.
Secure communication	Existing protocols break under quantum.	Replace with PQ communication protocols.

physical limitations like photon loss and decoherence. Advances in quantum error correction, repeaters, and entanglement swapping are necessary for long-distance secure communication.

Integrating PQC into blockchain and distributed ledger technologies will require rigorous mathematical modeling to maintain integrity and resilience against quantum threats. Overall, achieving practical, secure, and scalable quantum-safe systems requires continued research into cryptographic design, quantum network infrastructure, and global policy coordination.

In Table 16, we highlight the daunting challenges that quantum computing will introduce into the landscape of cybersecurity and discuss the paths to address these challenges. For example, algorithms such as Shor's and Grover's pose a threat to traditional cryptographic systems such as RSA and ECC; therefore, it is quite urgent to operate quantum-resistant cryptographic (PQC) algorithms, such as lattice-based algorithms or hash-based algorithms. The scalability problems in Quantum Key Distribution (QKD) and quantum networks need to be solved, and future solutions involve quantum repeaters and error correction, aiming to use these technologies for the extension of their applicability. Moreover, lightweight quantum-safe algorithms are required to protect resource-sparse IoT and edge devices against the same quantum threats.

9. Conclusion

The emergence of quantum computing represents a paradigm shift in cybersecurity, fundamentally challenging the foundations of modern cryptographic systems. Traditional encryption methods, including RSA and ECC, which have secured digital communications for decades, are now vulnerable to quantum attacks through Shor's and Grover's algorithms. These developments necessitate an immediate and comprehensive transition to quantum-resistant solutions to safeguard sensitive data, critical infrastructure, and digital economies.

Post-quantum cryptography (PQC) offers the most viable path forward, with lattice-based, hash-based, and multivariate cryptographic schemes demonstrating strong resistance to quantum attacks. However, their adoption faces significant challenges, including larger key sizes, computational overhead, and integration with legacy systems. Quantum key distribution (QKD) provides another layer of security through quantum mechanical principles, but its practical deployment is constrained by distance limitations and infrastructure requirements. Hybrid cryptographic systems, which combine classical and quantum-resistant algorithms, present a transitional solution, ensuring backward compatibility while future-proofing against quantum threats.

The transition to quantum-safe security extends beyond technical implementation. It requires global coordination among researchers, industry leaders, and policymakers to establish standardized protocols, optimize performance, and address scalability concerns. Critical sectors such as finance, healthcare, and government must prioritize quantum readiness, particularly for long-term data protection. Additionally, emerging technologies like blockchain and IoT demand tailored quantum-resistant solutions to mitigate risks in decentralized and resource-constrained environments.

While the timeline for large-scale quantum computing remains uncertain, the urgency for preparedness cannot be overstated. Proactive measures, including algorithm agility, continuous risk assessment, and workforce education, are essential to maintain robust cybersecurity in the quantum era. The journey toward quantum resilience is complex but imperative—a collaborative effort that will define the future of secure digital communication. By embracing innovation and fostering interdisciplinary cooperation, the global community can navigate this transformative shift and ensure a secure foundation for the next generation of technology.

Declaration of competing interest

The authors declare that they have no conflicts of interest related to this work. The research was conducted independently, without any financial or personal relationships that could influence the findings or interpretations presented in this paper.

Acknowledgments

The authors thank all the funding agencies and institutions that have supported this research. We also appreciate anonymous reviewers for their valuable feedback and constructive comments that greatly improved the quality of this paper. In addition to that, special thanks goes to colleagues and institutions for their assistance with data collection, analysis, and technical support. We are also very grateful to our families and friends for their enthusiastic support during the work.

Data availability

No data was used for the research described in the article.

References

- [1] Yazdi M. Application of quantum computing in reliability analysis. In: *Advances in computational mathematics for industrial system reliability and maintainability*. Springer; 2024, p. 139–54.
- [2] Alabi M. Quantum computing: A paradigm shift in computational power and its potential applications. 2024.
- [3] Li Z. Analysis of the principles of quantum computing and state-of-the-art applications. *Theor Nat Sci* 2024;41:65–71.
- [4] Khurana S, et al. Quantum machine learning: Unraveling a new paradigm in computational intelligence. *Quantum* 2024;74:1.
- [5] Lator C, Manssur L, Portugal R. Shor's algorithm for factoring large integers. 2003, arXiv preprint Quant-Ph/0303175.
- [6] Piron R, Habibie MI, Goursaud C. Mixed Grover: A hybrid version to improve Grover's algorithm for unstructured database search. *IEEE Trans Quantum Eng* 2025.
- [7] Akande B. The impact of quantum computing on encryption: How quantum computers can break current encryption methods, such as RSA and ECC, and what this means for data security. 2025.
- [8] Sarah D, Peter C. On the practical cost of Grover for AES key recovery. 2024.
- [9] Song G, Jang K, Kim H, Eum S, Sim M, Kim H, Lee W-K, Seo H. Grover on SPEEDY. *Cryptol EPrint Arch* 2021.
- [10] Battistel F, Chamberland C, Johar K, Overwater RW, Sebastiano F, Skoric L, Ueno Y, Usman M. Real-time decoding for fault-tolerant quantum computing: Progress, challenges and outlook. *Nano Futur* 2023;7(3):032003.
- [11] Takeda S, Furusawa A. Toward large-scale fault-tolerant universal photonic quantum computing. *APL Photonics* 2019;4(6).
- [12] Vasani V, Prateek K, Amin R, Maity S, Dwivedi AD. Embracing the quantum frontier: Investigating quantum communication, cryptography, applications and future directions. *J Ind Inf Integr* 2024;100594.
- [13] Ghosh D, Ghosh K, Chakraborty C, Datta A, Gupta S. Securing the future: Emerging threats and countermeasures in cryptography. *Secur Digit Front Threat Adv Tech in Secur Forensics* 2025;91–107.
- [14] Mamatha G, Dimri N, Sinha R. Post-quantum cryptography: Securing digital communication in the quantum era. 2024, arXiv preprint arXiv:2403.11741.
- [15] Alléaume R, Branciard C, Bouda J, Debuisschert T, Dianati M, Gisin N, Godfrey M, Grangier P, Länger T, Lütkenhaus N, et al. Using quantum key distribution for cryptographic purposes: A survey. *Theoret Comput Sci* 2014;560:62–81.
- [16] Awotidebe M. The race to quantum-resilient infrastructure: National and global strategies for cybersecurity readiness. 2025.
- [17] Bishwas AK, Sen M. Strategic roadmap for quantum-resistant security: A framework for preparing industries for the quantum threat. 2024, arXiv preprint arXiv:2411.09995.
- [18] Desai B, Patil K, Mehta I, Patil A. A secure communication framework for smart city infrastructure leveraging encryption, intrusion detection, and blockchain technology. *Adv Comput Sci* 2024;7(1).
- [19] Tsantikidou K, Sklavos N. Threats, attacks, and cryptography frameworks of cybersecurity in critical infrastructures. *Cryptography* 2024;8(1):7.
- [20] De Leon NP, Itoh KM, Kim D, Mehta KK, Northup TE, Paik H, Palmer B, Samarth N, Sangtawesin S, Steuerma DW. Materials challenges and opportunities for quantum computing hardware. *Science* 2021;372(6539):eabb2823.
- [21] Galindo A, Martin-Delgado MA. Information and computation: Classical and quantum aspects. *Rev Modern Phys* 2002;74(2):347.
- [22] Priyanka V, Reddy NS, Jeevana G, Arab MA. Design of arithmetic logic unit using reversible logic gates. In: 2024 2nd world conference on communication & computing. WCONF, IEEE; 2024, p. 1–6.
- [23] Hoffman KL, Padberg M, Rinaldi G, et al. Traveling salesman problem. *Encycl Oper Res Manag Sci* 2013;1:1573–8.
- [24] Cleri F. Quantum computers, quantum computing, and quantum thermodynamics. *Front Quantum Sci Technol* 2024;3:1422257.
- [25] Di Meglio A, Jansen K, Tavernelli I, Alexandrou C, Arunachalam S, Bauer CW, Borrás K, Carrazza S, Crippa A, Croft V, et al. Quantum computing for high-energy physics: State of the art and challenges. *PRX Quantum* 2024;5(3):037001.
- [26] Srivastava AK, Müller-Rigat G, Lewenstein M, Rajchel-Mieldzioć G. Introduction to quantum entanglement in many-body systems. In: *New trends and platforms for quantum technologies*. Springer; 2024, p. 225–85.
- [27] Markoska R, Markoski A. Quantum vs classical computing: Technologies in tandem. *Int J Recent Res Math Comput Sci Inf Technol* 2025;11(2).
- [28] Alrashed S, Min-Allah N. Quantum computing research in medical sciences. *Inform Med Unlocked* 2025;52:101606.
- [29] Mandelbaum R, Córcoles AD, Gambetta J. IBM's big bet on the quantum-centric supercomputer: Recent advances point the way to useful classical-quantum hybrids. *IEEE Spectr* 2024;61(9):24–33.
- [30] Gill SS, Buyya R. Transforming research with quantum computing. *J Econ Technol* 2024.
- [31] Abou El Houda Z, Moudoud H, Brik B, Adil M. A privacy-preserving framework for efficient network intrusion detection in consumer network using quantum federated learning. *IEEE Trans Consum Electron* 2024.
- [32] Eisert J, Wolf MM. Quantum computing. In: *Handbook of nature-inspired and innovative computing: Integrating classical models with emerging technologies*. Springer; 2006, p. 253–86.
- [33] Haldorai A. Advancements and applications of quantum computing in robotics. *J Comput Nat Sci* 2024;2(4):053–63.
- [34] Grabowska A, Gunia A. On quantum computing for artificial superintelligence. *Eur J Philos Sci* 2024;14(2):25.
- [35] Glassner A. An introduction to quantum computing. In: *ACM SIGGRAPH 2024 courses*. 2024, p. 1–65.
- [36] Youvan DC. The quantum architect: Exploring the implications of quantum interference as a designed feature of the universe. 2024.

- [37] Clenet M, Dion M, Blanchet FG. Addressing ecological challenges from a quantum computing perspective. 2025, arXiv preprint arXiv:2504.03866.
- [38] Ferry DK. Quantum information in the nanoelectronic world. 2024.
- [39] Memon QA, Al Ahmad M, Pecht M. Quantum computing: Navigating the future of computation, challenges, and technological breakthroughs. *Quantum Rep* 2024;6(4):627–63.
- [40] Akbar MA, Khan AA, Hyrynsalmi S. Role of quantum computing in shaping the future of 6 G technology. *Inf Softw Technol* 2024;170:107454.
- [41] Balewski J, Kornjača M, Klymko K, Darbha S, Hirsbrunner MR, Lopes PL, Liu F, Camps D. Engineering quantum states with neutral atoms. In: 2024 IEEE international conference on quantum computing and engineering. QCE, vol. 1, IEEE; 2024, p. 1221–7.
- [42] Balamurugan K, Sivakami A, Mathankumar M, Ahmad I, et al. Quantum computing basics, applications and future perspectives. *J Mol Struct* 2024;1308:137917.
- [43] Takook MV, Mohammad-Djafari A. Quantum states and quantum computing. 2024, arXiv preprint arXiv:2409.15285.
- [44] Paneru D, Cohen E, Fickler R, Boyd RW, Karimi E. Entanglement: Quantum or classical? *Rep Progr Phys* 2020;83(6):064001.
- [45] Caleffi M, Amoretti M, Ferrari D, Illiano J, Manzalini A, Cacciapuoti AS. Distributed quantum computing: A survey. *Comput Netw* 2024;254:110672.
- [46] Ekert A, Jozsa R. Quantum computation and Shor's factoring algorithm. *Rev Modern Phys* 1996;68(3):733.
- [47] Preskill J. Quantum computing and the entanglement frontier. 2012, arXiv preprint arXiv:1203.5813.
- [48] Bhat MI, Giri KJ. Impact of computational power on cryptography. *Multimed Secur Algorithm Dev Anal Appl* 2021;45–88.
- [49] Ambekar B, Bedi S. A new factorization method to factorize rsa public key encryption. *Int J Comput Sci Issues (IJCSI)* 2011;8(6):242.
- [50] Shatnawi AS, Almazari MM, AlShara Z, Taqieddin E, Mustafa D. RSA cryptanalysis—Fermat factorization exact bound and the role of integer sequences in factorization problem. *J Inf Secur Appl* 2023;78:103614.
- [51] Ugwuishiwu C, Orji U, Ugwu C, Asogwa C. An overview of quantum cryptography and Shor's algorithm. *Int J Adv Trends Comput Sci Eng.* 2020;9(5).
- [52] Nkulenu G. Quantum computing: The impending revolution in cryptographic security. 2024.
- [53] Yang D. Post-quantum cryptography: Effects of quantum computing on modern cryptography. *Int J High Sch Res* 2024;6(6).
- [54] Rayamajhi O. Enhancing Machine Learning Training Speed with Quantum Computing: A Comparative Analysis with Classical Computing.
- [55] Abiade O. Strategies for enhancing data encryption in the face of quantum computing threats. 2024.
- [56] Zhao X, Xu X, Qi L, Xia X, Bilal M, Gong W, Kou H. Unraveling quantum computing system architectures: An extensive survey of cutting-edge paradigms. *Inf Softw Technol* 2024;167:107380.
- [57] Kumar G, Yadav S, Mukherjee A, Hassija V, Guizani M. Recent advances in quantum computing for drug discovery and development. *IEEE Access* 2024;12:64491–509.
- [58] Gill SS, Cetinkaya O, Marrone S, Claudino D, Haunschild D, Schlote L, Wu H, Ottaviani C, Liu X, Machupalli SP, et al. Quantum computing: Vision and challenges. In: *Quantum computing*. Elsevier; 2025, p. 19–42.
- [59] Rieffel E, Polak W. An introduction to quantum computing for non-physicists. *ACM Comput Surv* 2000;32(3):300–35.
- [60] Shor PW. Introduction to quantum algorithms. In: *Proceedings of symposia in applied mathematics*. vol. 58, 2002, p. 143–60.
- [61] Junagade N, Bhandari SU, Patil RY. Comparative study of quantum algorithms: A comprehensive analysis. In: 2024 8th international conference on computing, communication, control and automation. ICCUBE, IEEE; 2024, p. 1–5.
- [62] Younis MM, Jamil AS, Abdulrazzaq AH, Mawla NA, Khudhair RM, Vasiliu Y. Progress and challenges in quantum computing algorithms for NP-hard problems. In: 2024 36th conference of open innovations association. FRUCT, IEEE; 2024, p. 460–8.
- [63] Souza V, Shetty N, Shetty S, Jison V. A systematic review on quantum computing: Concepts, algorithms, hardware and applications. *Algorithms Hardw Appl* 2024. (December 10, 2024).
- [64] Ghadi DM. Improving the robustness of RSA encryption through input-based key generation. *Math Model Eng Probl* 2024;11(1).
- [65] Arockia Christopher A, Sivakumar M, Muthanantha Murugavel A, Priyatharsini S, Angeljuliee J, Revathi P. Refined RSA encryption algorithm to safeguard privacy in healthcare data. In: *International conference on innovations in cybersecurity and data science proceedings of ICICDS*. Springer; 2024, p. 271–82.
- [66] Zhang K. The mathematical principles of RSA encryption algorithm. In: 2024 3rd international conference on artificial intelligence and computer information technology. AICIT, IEEE; 2024, p. 1–5.
- [67] Mishra S, Agarwal O, Patel SK. Quantum decryption using Shor's algorithm. In: 2024 IEEE pune section international conference. puneCon, IEEE; 2024, p. 1–6.
- [68] Thamilarasi V, Naik PK, Sharma I, Porkodi V, Sivaram M, Lawanyashri M. Quantum computing-navigating the frontier with Shor's algorithm and quantum cryptography. In: 2024 international conference on trends in quantum computing and emerging business technologies. IEEE; 2024, p. 1–5.
- [69] Sood N. Cryptography in post quantum computing era. 2024, Available At SSRN 4705470.
- [70] Firnhaber DK. Post-quantum cryptography: Constructive research in quantum simulations (Ph.D. thesis), Northcentral University; 2023.
- [71] Cai J-Y. Shor's algorithm does not factor large integers in the presence of noise. *Sci China Inf Sci* 2024;67(7):173501.
- [72] Wang W-L, Hussain S, Rondinelli NV, Tang M-H, Khan SUR. Recursive generation of multi-qubits control gate to facilitate the grover's quantum algorithm in large scale database search. In: 2024 IEEE 24th international conference on software quality, reliability, and security companion. QRS-c, IEEE; 2024, p. 808–15.
- [73] Yadav SK. Some problems in symmetric and asymmetric cryptography. LAP LAMBERT Academic Publishing; 2010.
- [74] Prokop M, Wallden P, Joseph D. Grover's oracle for the Shortest Vector Problem and its application in hybrid classical-quantum solvers. 2024, arXiv preprint arXiv:2402.13895.
- [75] Acampora G, Schiattarella R, Vitiello A. Using quantum amplitude amplification in genetic algorithms. *Expert Syst Appl* 2022;209:118203.
- [76] Al-Bayaty A, Perkowski M. A concept of controlling Grover diffusion operator: A new approach to solve arbitrary Boolean-based problems. *Sci Rep* 2024;14(1):23570.
- [77] Jain A, Praveen R, Musale V, Chintham N, Kumar Y, RamaKrishna B, Shrivastava A. Quantum computing and its implications for cryptography: Assessing the security and efficiency of quantum algorithms. *Libr Prog Int* 2024;44(3):5654–63.
- [78] Sharma M, Choudhary V, Bhatia R, Malik S, Raina A, Khandelwal H. Leveraging the power of quantum computing for breaking RSA encryption. *Cyber-Phys Syst* 2021;7(2):73–92.
- [79] Karakaya A, Ulu A. A survey on post-quantum based approaches for edge computing security. *Wiley Interdiscip Rev Comput Stat* 2024;16(1):e1644.
- [80] Malhotra J, Chavan O, Thakur P. Grover algorithm: Enhancing the search mechanism. In: 2025 4th OPJU international technology conference (OTCON) on smart computing for innovation and advancement in industry 5.0. IEEE; 2025, p. 1–8.
- [81] Srivastava S, Pati AK, Chakrabarty I, Bhattacharya S. Using quantum switches to mitigate noise in Grover's search algorithm. *J Phys A* 2025;58(10):105304.
- [82] Jura S-A, Udrescu M. Quantum-enhanced weight optimization for neural networks using Grover's algorithm. 2025, arXiv preprint arXiv:2504.14568.
- [83] Sabrina F, Sohail S, Tariq UU. A review of post-quantum privacy preservation for IoMT using blockchain. 2024.
- [84] Gangwar S, Pachnanda V, Pundir T. The impact of quantum computing on present cryptography. 2024, Available At SSRN 5017957.
- [85] Castro J. Quantum algorithms-design and analysis: Investigating the design and analysis of quantum algorithms for solving computational problems with quantum speedup, including Shor's and Grover's algorithms. *Aust J Mach Learn Res Appl* 2024;4(1):249–59.
- [86] Pham TMP. Post-quantum attacks against symmetric-key cryptography systems. 2024.
- [87] Ahmed N. Quantum computing algorithms for integer factorization: A comparative analysis. *Mod Dyn Math Progress* 2024;1(1):6–9.
- [88] Chirra DR. Quantum-safe cryptography: New frontiers in securing post-quantum communication networks. *Int J Mach Learn Res Cybersecur Artif Intell* 2024;15(1):670–88.

- [89] Shor PW. Algorithms for quantum computation: Discrete logarithms and factoring. In: Proceedings 35th annual symposium on foundations of computer science. IEEE; 1994, p. 124–34.
- [90] Grover LK. From Schrödinger's equation to the quantum search algorithm. Am J Phys 2001;69(7):769–77.
- [91] Weinstein YS, Pravia M, Fortunato E, Lloyd S, Cory DG. Implementation of the quantum Fourier transform. Phys Rev Lett 2001;86(9):1889.
- [92] Mohammadbagherpoor H, Oh Y-H, Dreher P, Singh A, Yu X, Rindos AJ. An improved implementation approach for quantum phase estimation on quantum computers. In: 2019 IEEE international conference on rebooting computing. ICRC, IEEE; 2019, p. 1–9.
- [93] Georgescu IM, Ashhab S, Nori F. Quantum simulation. Rev Modern Phys 2014;86(1):153–85.
- [94] Aharonov D. Quantum computation. Annu Rev Comput Phys VI 1999;259–346.
- [95] Nurhadi AI, Syambas NR. Quantum key distribution (QKD) protocols: A survey. In: 2018 4th international conference on wireless and telematics. ICWT, IEEE; 2018, p. 1–5.
- [96] AbuGhanem M. Photonic quantum computers. 2024, arXiv preprint arXiv:2409.08229.
- [97] Chander S. The current landscape of quantum hardware development-an overview. Intersect Stanf J Sci Technol Soc 2024;17(3).
- [98] Córcoles AD, Kandala A, Javadi-Abhari A, McClure DT, Cross AW, Temme K, Nation PD, Steffen M, Gambetta JM. Challenges and opportunities of near-term quantum computing systems. Proc IEEE 2019;108(8):1338–52.
- [99] Singh M, Carlson AH. An introduction to quantum computing and its applications. Cyber Déf Rev 2024;9(2):73–92.
- [100] Das A, Palesi M, Kim J, Pande PP. Chip and package-scale interconnects for general-purpose, domain-specific and quantum computing systems-overview, challenges and opportunities. IEEE J Emerg Sel Top Circuits Syst 2024.
- [101] AbuGhanem M, Eleuch H. NISQ computers: A path to quantum supremacy. IEEE Access 2024.
- [102] Zwerver A, Krähenmann T, Watson T, Lampert L, George HC, Pillarisetty R, Bojarski S, Amin P, Amitonov S, Boter J, et al. Qubits made by advanced semiconductor manufacturing. Nat Electron 2022;5(3):184–90.
- [103] De Teresa JM. Nanoscale direct-write fabrication of superconducting devices for application in quantum technologies. Mater Quantum Technol 2023;3(1):013001.
- [104] Nakamura K, Ankerhold J. Gate operations for superconducting qubits and non-Markovianity: Fidelities, long-range time correlations, and suppression of decoherence. 2024, arXiv preprint arXiv:2402.18518.
- [105] McDermott R. Materials origins of decoherence in superconducting qubits. IEEE Trans Appl Supercond 2009;19(1):2–13.
- [106] Khan MA, Ghafoor S, Zaidi SMH, Khan H, Ahmad A. From quantum communication fundamentals to decoherence mitigation strategies: Addressing global quantum network challenges and projected applications. Heliyon 2024.
- [107] Brown KR, Chiaverini J, Sage JM, Häffner H. Materials challenges for trapped-ion quantum computers. Nat Rev Mater 2021;6(10):892–905.
- [108] Lu Y, Zhang S, Zhang K, Chen W, Shen Y, Zhang J, Zhang J-N, Kim K. Global entangling gates on arbitrary ion qubits. Nature 2019;572(7769):363–7.
- [109] Bruzewicz CD, Chiaverini J, McConnell R, Sage JM. Trapped-ion quantum computing: Progress and challenges. Appl Phys Rev 2019;6(2).
- [110] Foss-Feig M, Pagano G, Potter AC, Yao NY. Progress in trapped-ion quantum simulation. Annu Rev Condens Matter Phys 2024;16.
- [111] Monroe C, Kim J. Scaling the ion trap quantum processor. Science 2013;339(6124):1164–9.
- [112] Goyal RK. Exploring quantum materials & applications: A review. 2024, arXiv preprint arXiv:2404.17594.
- [113] Schuckert A, Crane E, Gorshkov AV, Hafezi M, Gullans MJ. Fermion-qubit fault-tolerant quantum computing. 2024, arXiv preprint arXiv:2411.08955.
- [114] Nayak C, Simon SH, Stern A, Freedman M, Das Sarma S. Non-abelian anyons and topological quantum computation. Rev Modern Phys 2008;80(3):1083–159.
- [115] Lahtinen V, Pachos J. A short introduction to topological quantum computation. SciPost Phys 2017;3(3):021.
- [116] Weber B, Fuhrer MS, Sheng X-L, Yang SA, Thomale R, Shamim S, Molenkamp LW, Cobden D, Pesin D, Zandvliet HJ, et al. 2024 roadmap for 2D topological insulators. J Phys Mater 2024;7(2):022501.
- [117] Suter D, Álvarez GA. Colloquium: Protecting quantum information against environmental noise. Rev Modern Phys 2016;88(4):041001.
- [118] Anferov A, Harvey SP, Wan F, Simon J, Schuster DI. Superconducting qubits above 20 GHz operating over 200 mK. PRX Quantum 2024;5(3):030347.
- [119] Acharya R, Aghababae-Beni L, Aleiner I, Andersen TI, Ansmann M, Arute F, Arya K, Asfaw A, Astrakhantsev N, Atalaya J, et al. Quantum error correction below the surface code threshold. 2024, arXiv preprint arXiv:2408.13687.
- [120] Thantharath P, Thantharath A. Q-CODA: Co-designing quantum codes and architectures for hardware-aware quantum error correction. In: International symposium on quantum sciences: Applications and challenges. Springer; 2023, p. 134–51.
- [121] Tanttu T, Lim WH, Huang JY, Dumoulin Stuyck N, Gilbert W, Su RY, Feng M, Cifuentes JD, Seedhouse AE, Seritan SK, et al. Assessment of the errors of high-fidelity two-qubit gates in silicon quantum dots. Nat Phys 2024;1–6.
- [122] Fingerhuth M, Babej T, Wittek P. Open source software in quantum computing. PLoS One 2018;13(12):e0208561.
- [123] Ball H, Biercuk MJ, Carvalho AR, Chen J, Hush M, De Castro LA, Li L, Liebermann PJ, Slatyer HJ, Edmunds C, et al. Software tools for quantum control: Improving quantum computer performance through noise and error suppression. Quantum Sci Technol 2021;6(4):044011.
- [124] Fürntratt H, Schnabl P, Krebs F, Unterberger R, Zeiner H. Towards higher abstraction levels in quantum computing. In: International conference on service-oriented computing. Springer; 2023, p. 162–73.
- [125] Heim B, Soeken M, Marshall S, Granade C, Roetteler M, Geller A, Troyer M, Svore K. Quantum programming languages. Nat Rev Phys 2020;2(12):709–22.
- [126] Gonzalez-Zalba M, De Franceschi S, Charbon E, Meunier T, Vinet M, Dzurak A. Scaling silicon-based quantum computing using CMOS technology. Nat Electron 2021;4(12):872–84.
- [127] Sepúlveda S, Cravero A, Fonseca G, Antonelli L. Systematic review on requirements engineering in quantum computing: Insights and future directions. Electronics 2024;13(15):2989.
- [128] Sonko S, Ibekwe KI, Ilojiana VI, Etukudoh EA, Fabuyide A. Quantum cryptography and US digital security: A comprehensive review: Investigating the potential of quantum technologies in creating unbreakable encryption and their future in national security. Comput Sci IT Res J 2024;5(2):390–414.
- [129] Ajala OA, Arinze CA, Ofofiele OC, Okoye CC, Daraojimba AI. Exploring and reviewing the potential of quantum computing in enhancing cybersecurity encryption methods. 2024.
- [130] Imam R, Areeb QM, Alturki A, Anwer F. Systematic and critical review of rsa based public key cryptographic schemes: Past and present status. IEEE Access 2021;9:155949–76.
- [131] Dalal YM, Supreeth S, Amuthabala K, Satheesha T, Asha P, Somanath S. Optimizing security: A comparative analysis of RSA, ECC, and DH algorithms. In: 2024 IEEE North Karnataka subsection flagship international conference. nKCon, IEEE; 2024, p. 1–6.
- [132] Gyasi-Nyarko D, Freeman E, Ujakpa MM, Ampomah W, Amoako SO. A systematic review of public key cryptography: Implementation, challenges and future opportunities. In: 2025 IST-africa conference. IST-africa, IEEE; 2025, p. 1–15.
- [133] Meftah MM, Sa'ad HHY, Al-Ashmoery Y, Saad A-MH, Sa'd AHY, Alwesabi K. A comparative analysis of cryptography algorithms in information security. In: 2024 10th international conference on computing, engineering and design. ICCED, IEEE; 2024, p. 1–6.
- [134] Ketha A. The evolution of cryptography and a contextual analysis of the major modern schemes. 2023.
- [135] Liu F, Zheng Z, Gong Z, Tian K, Zhang Y, Hu Z, Li J, Xu Q. A survey on lattice-based digital signature. Cybersecurity 2024;7(1):7.
- [136] Tosun T, Moradi A, Savas E. Exploiting the central reduction in lattice-based cryptography. IEEE Access 2024.
- [137] Zeng C, He D, Feng Q, Peng C, Luo M. The implementation of polynomial multiplication for lattice-based cryptography: A survey. J Inf Secur Appl 2024;83:103782.
- [138] Sendrier N. Code-based cryptography. In: Encyclopedia of cryptography, security and privacy. Springer; 2025, p. 372–3.

- [139] Mishra S, Mondal B, Jha RK. A survey on isogeny-based cryptographic protocols. *Wirel Netw* 2025;31(3):2993–3024.
- [140] Goubin L, Patarin J, Yang B-Y. Multivariate cryptography. In: *Encyclopedia of cryptography, security and privacy*. Springer; 2024, p. 1–6.
- [141] Jaiswal CK, Sharma A, Chaturvedi A. Challenge response protocol using multivariate polynomial cryptography. In: *2025 IEEE international students' conference on electrical, electronics and computer science*. SCECS, IEEE; 2025, p. 1–5.
- [142] Deshpande A, Nalwade A, Gutte VS, Patil DR. Journeying through securing digital communication: A comparative analysis from classical to post-quantum cryptography. In: *2024 IEEE international conference on blockchain and distributed systems security*. ICBDS, IEEE; 2024, p. 1–7.
- [143] Nyári N. The impact of quantum computing on IT security. *Biztonságtudományi Szle* 2021;3(4):25–37.
- [144] Tom JJ, Anebo NP, Onyekwelu BA, Wilfred A, Eyo R. Quantum computers and algorithms: A threat to classical cryptographic systems. *Int J Eng Adv Technol* 2023;12(5):25–38.
- [145] Arukala H. Factorization in cybersecurity: A dual role of defense and vulnerability in the age of quantum computing. 2024.
- [146] Larasati HT, Kim H. Quantum cryptanalysis landscape of Shor's algorithm for elliptic curve discrete logarithm problem. In: *Information security applications: 22nd international conference, WISA 2021, Jeju island, South Korea, August 11–13, 2021, revised selected papers 22*. Springer; 2021, p. 91–104.
- [147] Ullah S, Zheng J, Din N, Hussain MT, Ullah F, Yousaf M. Elliptic curve cryptography; applications, challenges, recent advances, and future trends: A comprehensive survey. *Comput Sci Rev* 2023;47:100530.
- [148] Huang Y, Su Z, Zhang F, Ding Y, Cheng R. Quantum algorithm for solving hyperelliptic curve discrete logarithm problem. *Quantum Inf Process* 2020;19(2):62.
- [149] Fang W, Chen W, Zhang W, Pei J, Gao W, Wang G. Digital signature scheme for information non-repudiation in blockchain: A state of the art review. *EURASIP J Wirel Commun Netw* 2020;2020:1–15.
- [150] Mumtaz M, Ping L. Forty years of attacks on the RSA cryptosystem: A brief survey. *J Discret Math Sci Cryptogr* 2019;22(1):9–29.
- [151] Spies T. Public key infrastructure. In: *Computer and information security handbook*. Elsevier; 2017, p. 691–711.
- [152] Yunakovsky SE, Kot M, Pozhar N, Nabokov D, Kudinov M, Guglya A, Kiktenko EO, Kolycheva E, Borisov A, Fedorov AK. Towards security recommendations for public-key infrastructures for production environments in the post-quantum era. *EPJ Quantum Technol* 2021;8(1):14.
- [153] Vermeer MJ, Heitzenrater C, Parker E, Moon A, Lumpkin D, Awan J. Evaluating cryptographic vulnerabilities created by quantum computing in industrial control systems. *J Crit Infrastruct Policy* 2023.
- [154] Kumar DD, Mukharjee JD, Reddy CVD, Rajagopal SM. Safe and secure communication using SSL/TLS. In: *2024 international conference on emerging smart computing and informatics*. ESCI, IEEE; 2024, p. 1–6.
- [155] Babu PR, Kumar SA, Reddy AG, Das AK. Quantum secure authentication and key agreement protocols for IoT-enabled applications: A comprehensive survey and open challenges. *Comput Sci Rev* 2024;54:100676.
- [156] Moriarty KM. Authentication and authorization. In: *Transforming information security*. Emerald Publishing Limited; 2020, p. 131–44.
- [157] Ganesan R. Yaksha: Augmenting Kerberos with public key cryptography. In: *Proceedings of the symposium on network and distributed system security*. IEEE; 1995, p. 132–43.
- [158] Ferry E, O. Raw J, Curran K. Security evaluation of the OAuth 2.0 framework. *Inf Comput Secur* 2015;23(1):73–101.
- [159] Harding P, Johansson L, Klingenstein N. Dynamic security assertion markup language: Simplifying single sign-on. *IEEE Secur Priv* 2008;6(2):83–5.
- [160] Lakhani AS. A comparative study on post-quantum cryptographic digital signature algorithms: Network performance, key robustness, and energy consumption (Ph.D. thesis), Carleton University; 2023.
- [161] Andreou A, Mavroumoustakis CX, Markakis EK, Mastorakis G, Pallis E, Bourdena A. Exploring quantum-resistant cryptography solutions for health data exchange. In: *Intelligent technologies for healthcare business applications*. Springer; 2024, p. 19–47.
- [162] Simmons GJ. Symmetric and asymmetric encryption. *ACM Comput Surv* 1979;11(4):305–30.
- [163] Pappachan P, Rahaman M, Sreerakuvandana S, Bansal S, Arya V. Beyond current cryptography: Exploring new frontiers. In: *Innovations in modern cryptography*. IGI Global; 2024, p. 1–30.
- [164] Hazra R, Chatterjee P, Singh Y, Podder G, Das T. Data encryption and secure communication protocols. In: *Strategies for e-commerce data security: Cloud, blockchain, AI, and machine learning*. IGI Global; 2024, p. 546–70.
- [165] Ali S, Wang J, Leung VCM. Defensive strategies against PCC attacks based on ideal (t, n) -secret sharing scheme. *J King Saud Univ-Comput Inf Sci* 2023;35(9):101784.
- [166] Ali S, Wadho SA, Yichiet A, Gan ML, Lee CK. Advancing cloud security: Unveiling the protective potential of homomorphic secret sharing in secure cloud computing. *Egypt Inform J* 2024;27:100519.
- [167] Wadho SA, Ali S, Mohammed AAA, Yichiet A, Gan ML, Lee CK. Secret sharing as a defense mechanism for ransomware in cloud storage systems. *Int J Adv Comput Sci Appl* 2024;15(10).
- [168] Shanta JV. Evaluating the performance of symmetric key algorithms: AES (advanced encryption standard) and DES (data encryption standard). *IJCEM Int J Comput Eng Manag* 2012;15(4):43–9.
- [169] Khan BUI, Olanrewaju RF, Morshidi MA, Mir RN, Kiah MLBM, Khan AM. Evolution and analysis of secure hash algorithm (Sha) family. *Malays J Comput Sci* 2022;35(3):179–200.
- [170] Kaur R, Kaur A. Digital signature. In: *2012 international conference on computing sciences*. IEEE; 2012, p. 295–301.
- [171] Möller M, Vuik C. On the impact of quantum computing technology on future developments in high-performance scientific computing. *Ethics Inf Technol* 2017;19:253–69.
- [172] Popa RA, Redfield CM, Zeldovich N, Balakrishnan H. CryptDB: Protecting confidentiality with encrypted query processing. In: *Proceedings of the twenty-third ACM symposium on operating systems principles*. 2011, p. 85–100.
- [173] Tariq U, Ahmed I, Bashir AK, Shaikat K. A critical cybersecurity analysis and future research directions for the internet of things: A comprehensive review. *Sensors* 2023;23(8):4117.
- [174] Singh B, Ahateshaam M, Lahiri A, Sagar AK. Future of cryptography in the era of quantum computing. In: *International conference on electrical and electronics engineering*. Springer; 2023, p. 13–31.
- [175] Gajbhiye S, Karmakar S, Sharma M, Sharma S. Paradigm shift from classical cryptography to quantum cryptography. In: *2017 international conference on intelligent sustainable systems*. ICISS, IEEE; 2017, p. 548–55.
- [176] Baseri Y, Chouhan V, Hafid A. Navigating quantum security risks in networked environments: A comprehensive study of quantum-safe network protocols. *Comput Secur* 2024;142:103883.
- [177] Naik AS, Yeniaras E, Hellstern G, Prasad G, Vishwakarma SKLP. From portfolio optimization to quantum blockchain and security: A systematic review of quantum computing in finance. *Financ Innov* 2025;11(1):1–67.
- [178] Radanliev P. Cyber diplomacy: Defining the opportunities for cybersecurity and risks from artificial intelligence, IoT, blockchains, and quantum computing. *J Cyber Secur Technol* 2025;9(1):28–78.
- [179] Sivan A, Priya K. Quantum computing and risk prediction accuracy: An analysis of IT companies' risk appetite. *Int J Bus Syst Res* 2025;19(2):111–39.
- [180] Rishiwal V, Agarwal U, Yadav M, Tanwar S, Garg D, Guizani M. A new alliance of machine learning and quantum computing: Concepts, attacks, and challenges in iot networks. *IEEE Internet Things J* 2025.
- [181] Rattan A, Pal AR, Gurusamy M. Quantum computing for advanced driver assistance systems and autonomous vehicles: A review. *IEEE Access* 2025.
- [182] Wicaksana A. A survey on quantum-safe blockchain security infrastructure. *Comput Sci Rev* 2025;57:100752.

- [183] Larocca M, Thanasilp S, Wang S, Sharma K, Biamonte J, Coles PJ, Cincio L, McClean JR, Holmes Z, Cerezo M. Barren plateaus in variational quantum computing. *Nat Rev Phys* 2025;1–16.
- [184] Baklaga L. Quantum-resistant lattice-based cryptography: New conjectures on the learning with errors problem. *Sci Pr Cyber Secur J* 2024.
- [185] Xu D, Wang X, Hao Y, Zhang Z, Hao Q, Zhou Z. A more accurate and robust binary ring-LWE decryption scheme and its hardware implementation for IoT devices. *IEEE Trans Very Large Scale Integr (VLSI) Syst* 2022;30(8):1007–19.
- [186] Albrecht M, Cid C, Faugere J-C, Fitzpatrick R, Perret L. Algebraic algorithms for LWE problems. 2014.
- [187] Balamurugan C, Singh K, Ganesan G, Rajarajan M. Post-quantum and code-based cryptography—some prospective research directions. *Cryptography* 2021;5(4):38.
- [188] Tambe-Jagtap SN. A survey of cryptographic algorithms in cybersecurity: From classical methods to quantum-resistant solutions. *SHIFRA* 2023;2023:43–52.
- [189] Srivastava V, Debnath SK, Tiwari SK, Singh H. On the security of multivariate-based ring signature and other related primitives. *J Inf Secur Appl* 2023;74:103474.
- [190] Fathalla E, Azab M. Beyond classical cryptography: A systematic review of post-quantum hash-based signature schemes, security, and optimizations. *IEEE Access* 2024.
- [191] Nejatollahi H, Dutt N, Ray S, Regazzoni F, Banerjee I, Cammarota R. Post-quantum lattice-based cryptography implementations: A survey. *ACM Comput Surv* 2019;51(6):1–41.
- [192] Señor J, Portilla J, Mujica G. Analysis of the ntru post-quantum cryptographic scheme in constrained iot edge devices. *IEEE Internet Things J* 2022;9(19):18778–90.
- [193] Ugbedeojio M, Adebisi MO, Aroba OJ, Adebisi AA. RSA and elliptic curve encryption system: A systematic literature review. *Int J Inf Secur Priv (IJISP)* 2024;18(1):1–27.
- [194] Obert J, Cordeiro P, Johnson JT, Lum G, Tansy T, Pala N, Ih R. Recommendations for trust and encryption in DER interoperability standards. Tech. rep., Sandia National Lab.(SNL-NM), Albuquerque, NM (United States); Kitu Systems ...; 2019.
- [195] Rusca D, Gisin N. Quantum cryptography: An overview of quantum key distribution. 2024, arXiv preprint arXiv:2411.04044.
- [196] Cao Y, Zhao Y, Wang Q, Zhang J, Ng SX, Hanzo L. The evolution of quantum key distribution networks: On the road to the qinternet. *IEEE Commun Surv Tutor* 2022;24(2):839–94.
- [197] Surla G, Lakshmi R. Design and evaluation of novel hybrid quantum resistant cryptographic system for enhancing security in wireless body sensor networks. *Opt Quantum Electron* 2023;55(14):1252.
- [198] Nisha F, Lenin J, Saravanan S, Rohit VR, Selvam P, Rajmohan M. Lattice-based cryptography and NTRU: Quantum-resistant encryption algorithms. In: 2024 international conference on emerging systems and intelligent computing. ESIC, IEEE; 2024, p. 509–14.
- [199] Vadisetty R, Polamarasetti A. Quantum computing for cryptographic security with artificial intelligence. In: 2024 12th international conference on control, mechatronics and automation. ICCMA, IEEE; 2024, p. 252–60.
- [200] Wu F, Zhou B, Song J, Xie L. Quantum-resistant blockchain and performance analysis. *J Supercomput* 2025;81(3):498.
- [201] Khan MA, Javaid S, Mohsan SAH, Tanveer M, Ullah I. Future-proofing security for UAVs with post-quantum cryptography: A review. *IEEE Open J Commun Soc* 2024.
- [202] Wolf C. Multivariate quadratic polynomials in public key cryptography. *Cryptol EPrint Arch* 2005.
- [203] Khodaiehmehr H, Bagheri K, Feng C. Navigating the quantum computing threat landscape for blockchains: A comprehensive survey. 2023, Authorea Preprints.
- [204] Ankunda PV. Future-proofing medical devices: A hybrid approach to security in the post-quantum computing era (Ph.D. thesis), California State Polytechnic University, Pomona; 2023.
- [205] Alhakami H. Enhancing IoT security: Quantum-level resilience against threats. *Comput Mater Contin* 2024;78(1).
- [206] Scholten TL, Williams CJ, Moody D, Mosca M, Hurley W, Zeng WJ, Troyer M, Gambetta JM, et al. Assessing the benefits and risks of quantum computers. 2024, arXiv preprint arXiv:2401.16317.
- [207] Aydeger A, Zeydan E, Yadav AK, Hemachandra KT, Liyanage M. Towards a quantum-resilient future: Strategies for transitioning to post-quantum cryptography. In: 2024 15th international conference on network of the future. noF, IEEE; 2024, p. 195–203.
- [208] Sridhar S, Smys S. Hybrid RSAECC based secure communication in mobile cloud environment. *Wirel Pers Commun* 2020;111(1):429–42.
- [209] Gharavi H, Granjal J, Monteiro E. Post-quantum blockchain security for the internet of things: Survey and research directions. *IEEE Commun Surv Tutor* 2024.
- [210] Dhinakaran D, Selvaraj D, Dharini N, Raja SE, Priya C. Towards a novel privacy-preserving distributed multiparty data outsourcing scheme for cloud computing with quantum key distribution. 2024, arXiv preprint arXiv:2407.18923.
- [211] Sidhu JS, Joshi SK, Gündoğan M, Brougham T, Lowndes D, Mazzarella L, Krutzik M, Mohapatra S, Dequal D, Vallone G, et al. Advances in space quantum communications. *IET Quantum Commun* 2021;2(4):182–217.
- [212] Fernández-Caramés TM. From pre-quantum to post-quantum IoT security: A survey on quantum-resistant cryptosystems for the Internet of Things. *IEEE Internet Things J* 2019;7(7):6457–80.
- [213] Kan K, Une M, et al. Recent trends on research and development of quantum computers and standardization of post-quantum cryptography. 2021.
- [214] Chawla D, Mehra PS. A roadmap from classical cryptography to post-quantum resistant cryptography for 5G-enabled IoT: Challenges, opportunities and solutions. *Internet Things* 2023;100950.
- [215] Sikiru IA, Kora AD, Ezin EC, Imoize AL, Li C-T. Hybridization of learning techniques and quantum mechanism for IIoT security: Applications, challenges, and prospects. *Electronics* 2024;13(21):4153.
- [216] Tariq L, Atta A, Farooq U, Anwar N, Asim M, Tabassum N. Quantum-inspired cryptography protocols for enhancing security in cloud computing infrastructures. *Stat Comput Interdiscip Res* 2024;6(1):19–31.
- [217] Faridi AR, Masood F, Shamsan AHT, Luqman M, Salmony MY. Blockchain in the quantum world. 2022, arXiv preprint arXiv:2202.00224.
- [218] Singh MP, Singh J, Ravi V, Gupta P, Alahmadi TJ, Singh P, Shivahare BD, Verma M. Impact and implications of quantum computing on blockchain-based electronic health record systems. *Open Bioinform J* 2024;17(1).
- [219] Ravi P, Howe J, Chattopadhyay A, Bhasin S. Lattice-based key-sharing schemes: A survey. *ACM Comput Surv* 2021;54(1):1–39.
- [220] Srivastava V, Baksi A, Debnath SK. An overview of hash based signatures. *Cryptol EPrint Arch* 2023.
- [221] De Oliveira AKD, Cabral R, et al. High performance of hash-based signature schemes. *Int J Adv Comput Sci Appl* 2017;8(3).
- [222] Yokubov B. Post-quantum blockchain for internet of things domain (Ph.D. thesis), Brunel University London; 2023.
- [223] Sharma P, Choi K, Krejcar O, Blazek P, Bhatia V, Prakash S. Securing optical networks using quantum-secured blockchain: An overview. *Sensors* 2023;23(3):1228.
- [224] Rath S, Kundu NK, Sahoo S. Dynamic quantum key distribution for microgrids with distributed error correction. 2024, arXiv preprint arXiv:2405.11245.
- [225] Suhail S, Hussain R, Khan A, Hong CS. On the role of hash-based signatures in quantum-safe internet of things: Current solutions and future directions. *IEEE Internet Things J* 2020;8(1):1–17.
- [226] Lloyd-Jones S, Manwaring K. Quantum resilience in the Australian national security legislative framework (policy brief). 2024, Available At SSRN 4936141.
- [227] Namasudra S, Roy P. A new secure authentication scheme for cloud computing environment. *Concurr Comput Pr Exp* 2017;29(20):e3864.

- [228] Das S, Namasudra S. A lightweight and anonymous mutual authentication scheme for medical big data in distributed smart healthcare systems. *IEEE/ACM Trans Comput Biol. Bioinform* 2022.
- [229] Kumar T, Kumar P, Namasudra S. User revocation-enabled access control model using identity-based signature in the cloud computing environment. 2024.
- [230] Javeed D, Saeed MS, Ahmad I, Adil M, Kumar P, Islam AN. Quantum-empowered federated learning and 6G wireless networks for IoT security: Concept, challenges and future directions. *Future Gener Comput Syst* 2024.
- [231] Ahmed S, Khan M. Securing the Internet of Things (IoT): A comprehensive study on the intersection of cybersecurity, privacy, and connectivity in the IoT ecosystem. *AI IoT Fourth Ind Revolut Rev* 2023;13(9):1–17.
- [232] Ali S, Wang J, Leung VCM, Ali A. Decentralized ransomware recovery network: Enhancing resilience and security through secret sharing schemes. In: *IoTBDs*. 2024, p. 294–301.
- [233] Shannon K, Towe E, Tonguz OK. On the use of quantum entanglement in secure communications: A survey. 2020, arXiv preprint [arXiv:2003.07907](https://arxiv.org/abs/2003.07907).
- [234] Kong L, Tan J, Huang J, Chen G, Wang S, Jin X, Zeng P, Khan M, Das SK. Edge-computing-driven internet of things: A survey. *ACM Comput Surv* 2022;55(8):1–41.
- [235] Majdoubi C, El Mendili S, Gahi Y. Quantum cryptology in the big data security era. *Quantum* 2024;15(7).
- [236] Adnan MH, Ahmad Zukarnain Z, Harun NZ. Quantum key distribution for 5g networks: A review, state of art and future directions. *Futur Internet* 2022;14(3):73.
- [237] Mehic M, Niemiec M, Rass S, Ma J, Peev M, Aguado A, Martin V, Schauer S, Poppe A, Pacher C, et al. Quantum key distribution: A networking perspective. *ACM Comput Surv* 2020;53(5):1–41.
- [238] Rane J, Mallick S, Kaya O, Rane N. Artificial intelligence, machine learning, and deep learning in cloud, edge, and quantum computing: A review of trends, challenges, and future directions. *Futur Res Oppor Artif Intell Ind* 4.0 2024;5:2.
- [239] Singh A, Dev K, Siljak H, Joshi HD, Magarini M. Quantum internet—applications, functionalities, enabling technologies, challenges, and research directions. *IEEE Commun Surv Tutor* 2021;23(4):2218–47.
- [240] Ghonaimy MA. An overview of quantum information systems. In: 2013 8th international conference on computer engineering & systems. ICCES, IEEE; 2013, p. xx–xxxii.
- [241] Turci A. Advancements in Quantum Key Distribution: Achieving Secure Communication.
- [242] Pirandola S, Eisert J, Weedbrook C, Furusawa A, Braunstein SL. Advances in quantum teleportation. *Nat Photonics* 2015;9(10):641–52.
- [243] Nourmandipour A, Tavassoly M. Entanglement swapping between dissipative systems. *Phys Rev A* 2016;94(2):022339.
- [244] Azuma K, Economou SE, Elkouss D, Hilaire P, Jiang L, Lo H-K, Tzitrin I. Quantum repeaters: From quantum networks to the quantum internet. *Rev Modern Phys* 2023;95(4):045006.
- [245] Bhuyan AK, Dutta H. Harnessing quantum entanglement: Comprehensive strategies for enhanced communication and beyond in quantum networks. 2024, arXiv preprint [arXiv:2406.08833](https://arxiv.org/abs/2406.08833).
- [246] Baseri Y, Chouhan V, Ghorbani A. Cybersecurity in the quantum era: Assessing the impact of quantum computing on infrastructure. 2024, arXiv preprint [arXiv:2404.10659](https://arxiv.org/abs/2404.10659).
- [247] Jowarder RA, Jahan S. Quantum computing in cyber security: Emerging threats, mitigation strategies, and future implications for data protection. *World J Adv Eng Technol Sci* 2024;13(1):330–9.
- [248] Mangla C, Rani S, Qureshi NMF, Singh A. Mitigating 5G security challenges for next-gen industry using quantum computing. *J King Saud Univ-Comput Inf Sci* 2023;35(6):101334.
- [249] Faruk MJH, Tahora S, Tasnim M, Shahriar H, Sakib N. A review of quantum cybersecurity: Threats, risks and opportunities. In: 2022 1st international conference on AI in cybersecurity. ICAIC, IEEE; 2022, p. 1–8.
- [250] Moura LFAd, Westphall CB. A tertiary review on quantum cryptography. 2025, arXiv preprint [arXiv:2506.02028](https://arxiv.org/abs/2506.02028).
- [251] Ahmed U, Sipola T, Hautamäki J. Cyber protection applications of quantum computing: A review. 2024, arXiv preprint [arXiv:2406.13259](https://arxiv.org/abs/2406.13259).

Sijjad Ali is currently pursuing a PhD at Shenzhen University in the College of Computer Science and Software Engineering. He has published several research articles in well-reputed conferences and journals. His research interests include information security, cloud security, secret sharing schemes, and artificial intelligence.

Shuaib Ahmed Wadho holds degrees from the University of Sindh, Quaid-i-Azam University, and SZABIST. He is pursuing a Ph.D. in Computer Science at UTAR, Malaysia. His research focuses on cybersecurity, security assurance, cloud security, artificial intelligence, machine learning, and deep learning.

Kazim Raza Talpur is a Senior Lecturer at Universiti Teknologi Malaysia. He holds a Ph.D. in Computer Science from UTM. His research interests include text mining, machine learning, natural language processing, and online social networks.

Bandeh Ali Talpur is a Ph.D. candidate at Trinity College Dublin and a Senior Data Engineer at HSE, Ireland. With 8+ years of experience, his research focuses on NLP and machine learning, particularly hate speech, cybercrime, and cyberbullying detection in online social networks.

Khulud Salem Alshudukhi is Assistant Professor at Jouf University, specializes in deep learning, machine learning, cybersecurity, privacy, IoT, and blockchain. She actively contributes to research and peer reviews, focusing on applying emerging technologies to real-world security challenges in smart environments and critical sectors.

Humayun Mamoon is a Senior Lecturer at the University of Roehampton, London. She has 250+ publications, international patents, and extensive research in AI, cybersecurity, IoT, and software engineering. She also supervises postgraduates and serves as a reviewer and keynote speaker globally.

Samar Raza Talpur currently serves as the Director of ICT/Director of the Khairpur/Kandhkot Campuses at IBA Sukkur University. He earned his Ph.D. in Computer Science from University College Dublin (UCD), Dublin, Ireland, in 2016. His primary research focuses on networking and wireless networking, particularly in detecting hate speech, cybercrime, and cyberbullying behavior within online social networks.

Mohd Abdullah Al Mamun is pursuing a Master's in Information Technology Management at Westcliff University, USA. His interests include cybersecurity, cloud infrastructure, digital transformation, and IT governance. He is actively engaged in research to contribute to information systems and technology management through scholarly work.

Muhammad Naseem is pursuing a PhD in the College of Media and Communication at Shenzhen University, China. His research focuses on political communication and artificial intelligence, exploring how emerging technologies influence media, public opinion, and political discourse in contemporary digital environments.

Adeel Abro is Head of IT at SABIS University, Jamshoro, holds a Ph.D. from Beijing University of Posts and Telecommunications. With 20 years' experience, he specializes in AI, cybersecurity, leads national projects, and has received awards for innovative IT and smart classroom initiatives.

Dhani Bux Talpur is Assistant Professor at the University of Sufism and Modern Sciences, holds a Ph.D. from Guilin University, China. He specializes in machine learning and NLP, focuses on sentiment analysis, has leadership experience, and is an accomplished researcher and CSC scholar.

Asadullah Shah hold Ph.D. in Multimedia Communication, is a senior academic at IIUM with 180+ publications, extensive teaching and leadership experience, and expertise in cloud computing, e-learning, and machine learning. He supervises Ph.D. students and leads IIUM Press.